🏆 Infosys SE Elite Crack Vault — 10 College ITM Reasoning Ability 105Q Exam Compendium Elite

🎯 Infosys Systems Engineer (SE) | 📊 73 Questions | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

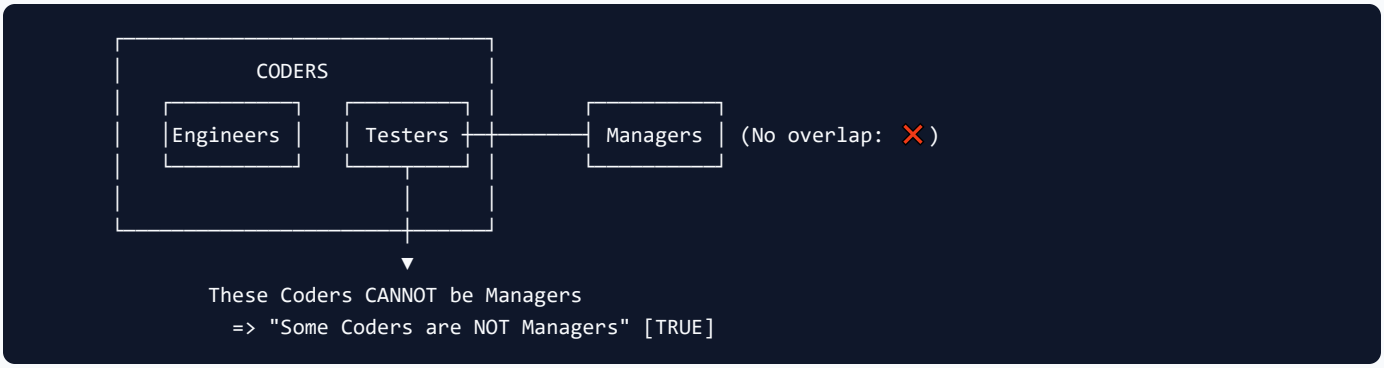
Problem 1 • Infosys SE Question 1 MUST-SOLVE

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

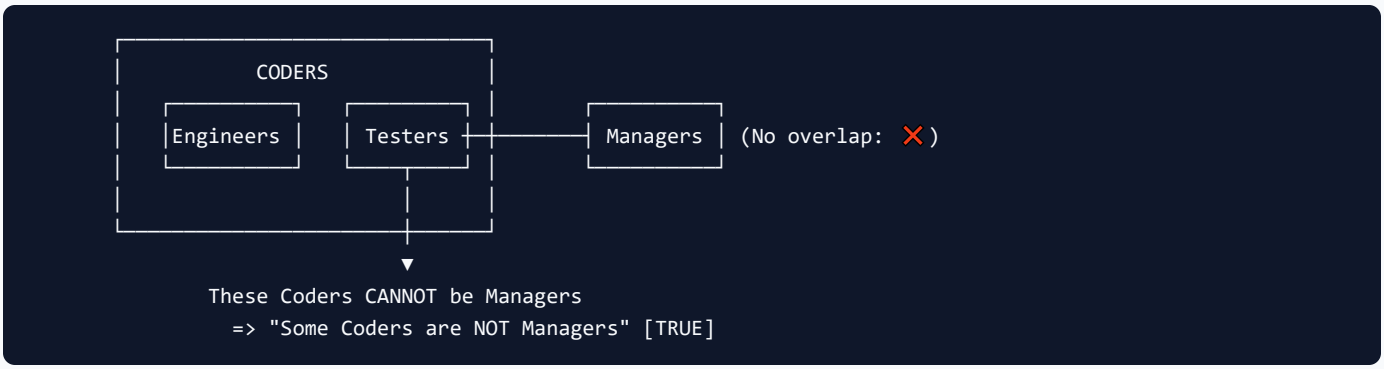
(Infosys System Engineer ITM Mock Paper 1 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_01_page_01.png

Question Text:
Statements: All engineers are coders. Some coders are testers. No tester is a manager.
Conclusions: I. Some coders are not managers. II. Some engineers are testers. III. No engineer is a manager.

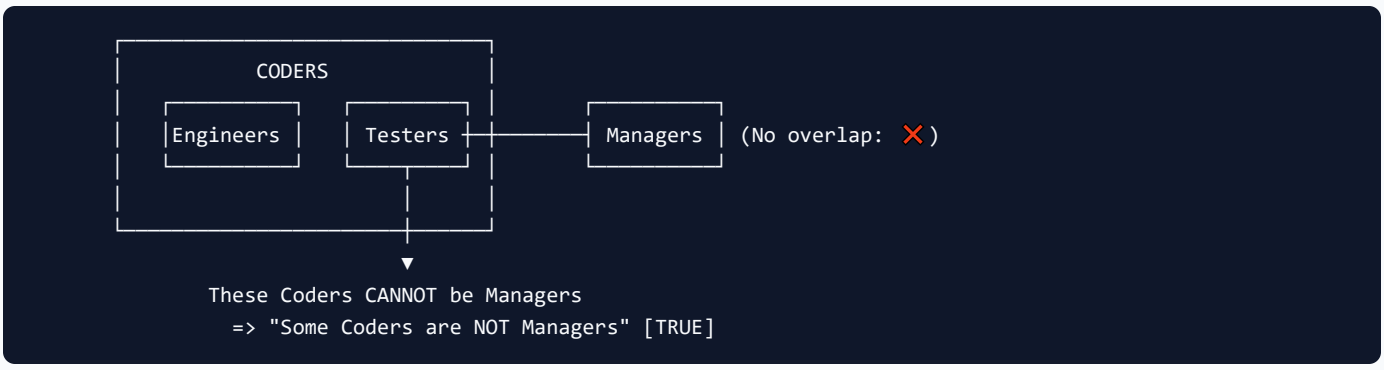
Visual Schematic / Logical Deduction Diagram:



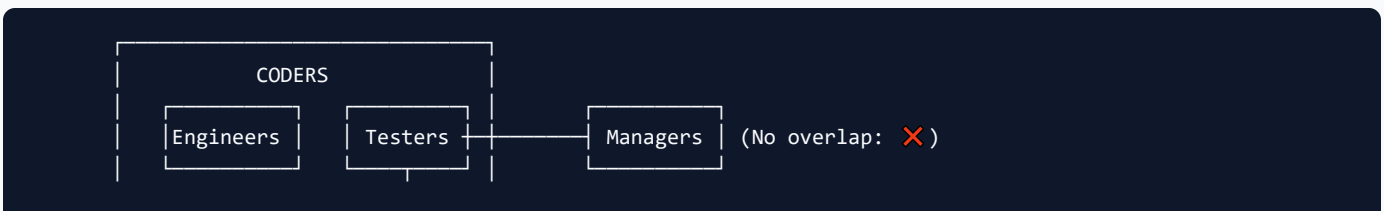
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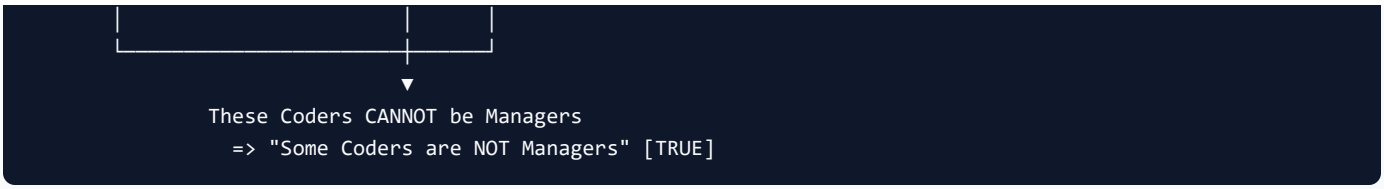


Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:





A Only I follows

B I and II follow

C I and III follow

D All follow

✓ **Verified Answer:** (A)** — `Only I follows` **Detailed Step-by-Step Solution & Logical Derivation:** Statement I: All engineers are coders, Some coders are testers, No tester is a manager. The subset of coders who are testers cannot be managers; hence 'Some coders are not managers' follows definitively. II (Some engineers are testers) does not necessarily follow as the circle of engineers may not overlap with testers. III (No engineer is a manager) does not necessarily follow. Thus, only conclusion I follows.

Problem 2 • Infosys SE Question 2

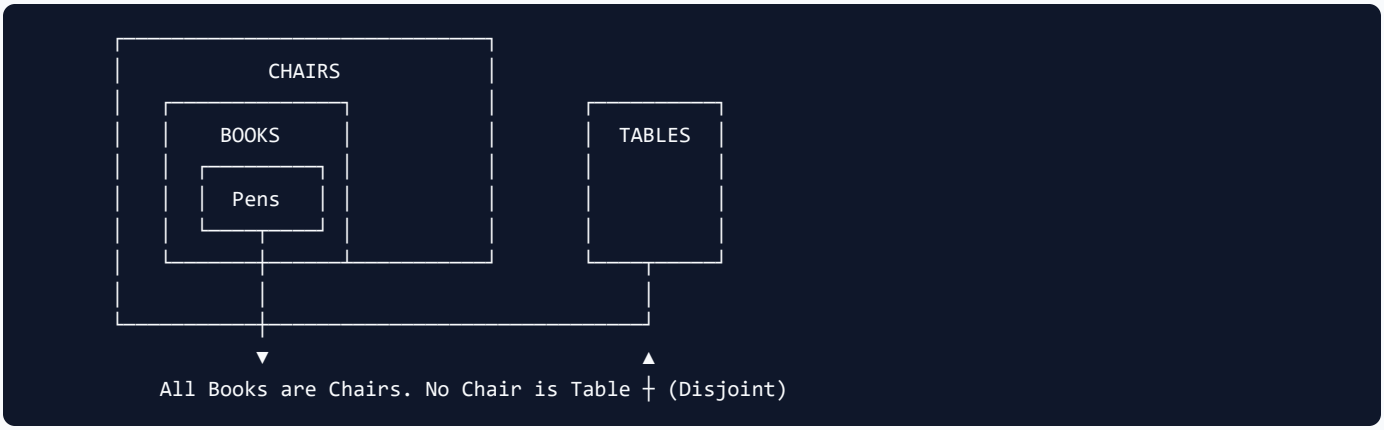
MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

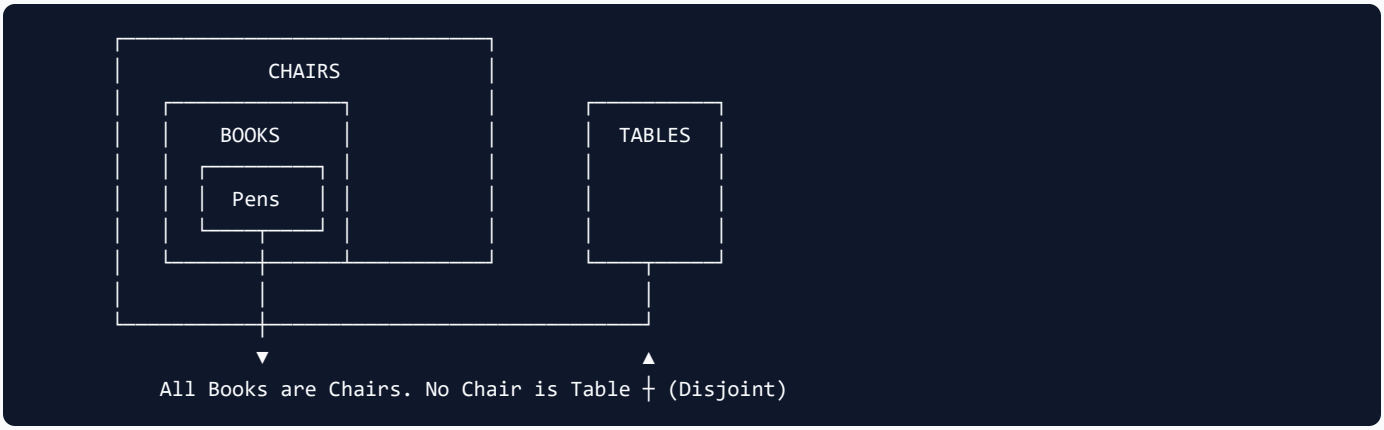
(Infosys System Engineer ITM Mock Paper 1 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_01_page_01.png

Question Text:
Statements: Some pens are books. All books are chairs. No chair is a table.
Conclusions: I. Some pens are chairs. II. No book is a table. III. Some pens are not tables. IV. Some tables are pens.

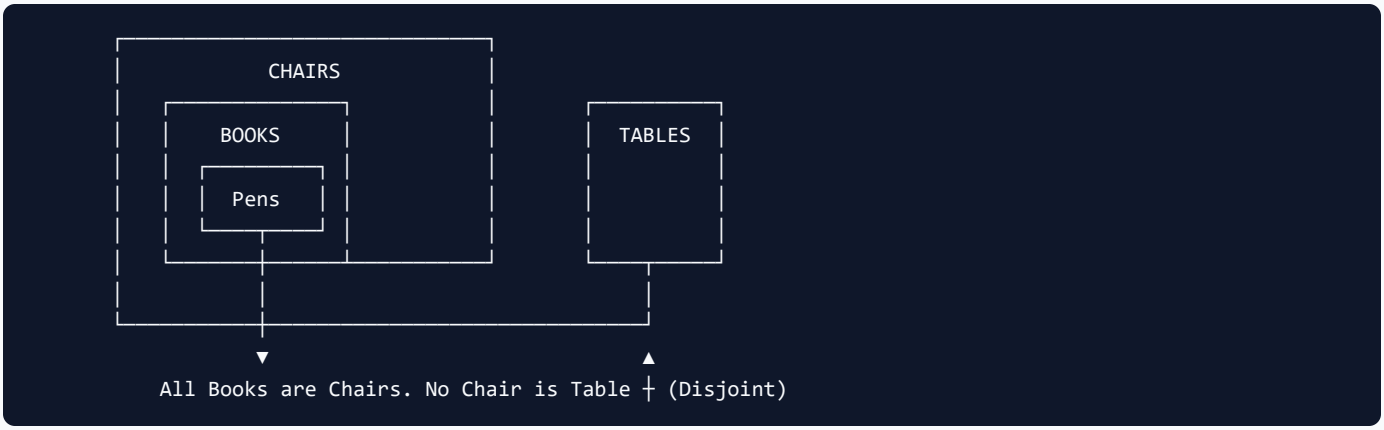
Visual Schematic / Logical Deduction Diagram:



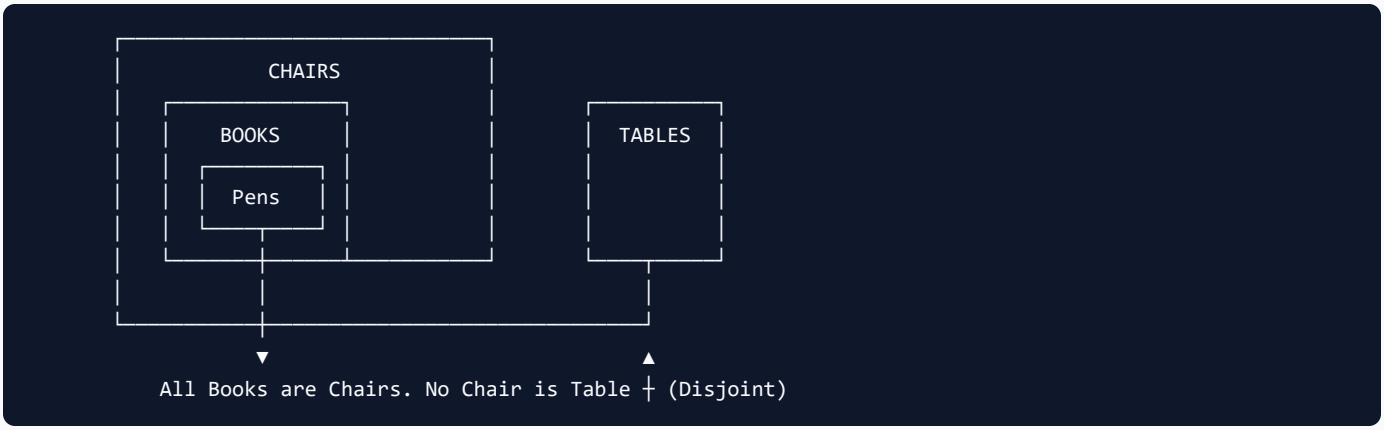
Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



A I and II only

B I, II and III only

C II, III and IV only

D All follow

✓ **Verified Answer:** (B)** — `I, II and III only` **Detailed Step-by-Step Solution & Logical Derivation:** Statements: Some pens are books. All books are chairs. No chair is a table. - Conclusion I: Some pens are chairs. Since some pens are books and all books are chairs, pens overlapping books must also overlap chairs. (Follows) - Conclusion II: No book is a table. Since all books are inside chairs and no chair is a table, books cannot overlap tables. (Follows) - Conclusion III: Some pens are not tables. The pens that are chairs cannot be tables. (Follows) - Conclusion IV: Some tables are pens. No positive overlap between tables and pens is guaranteed. (Does not follow) Therefore, I, II and III only follow.

Problem 3 • Infosys SE Question 3

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_01_page_01.png

Question Text:
Data Sufficiency. For integers x and y, is x > y?
I. $x^2 > y^2$. II. x + y > 0.

A. I alone is sufficient

B. II alone is sufficient

C. Both together are sufficient, neither alone

D. Even both

✓ **Verified Answer:** (C)** — `Both together are sufficient, neither alone` **Detailed Step-by-Step Solution & Logical Derivation:** Question: Is x > y for integers x and y? - Statement I: $x^2 > y^2$ gives |x| > |y|. If x = -3, y = 2, $x^2 = 9 > 4$ but x < y. If x = 3, y = 2, x > y. Insufficient. - Statement II: x + y > 0 gives x > -y. If x = -1, y = 3, x < y. If x = 3, y = -1, x > y. Insufficient. - Both together: $x^2 > y^2$ => $x^2 - y^2 > 0$ => (x - y)(x + y) > 0. Since Statement II guarantees (x + y) > 0, we can safely divide both sides by (x + y) without changing the inequality sign, yielding (x - y) > 0, which means x > y! Definite YES. Both together are sufficient, neither alone.

Problem 4 • Infosys SE Question 4

MUST-SOLVE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_01_page_01.png

Question Text:
Data Sufficiency. What is Riya's present age?
I. Riya is twice as old as her sister, who is 3 years younger than their brother.
II. Five years hence, the sum of the ages of Riya and her sister will be 40.

A I alone is sufficient

B II alone is sufficient

C Both together are sufficient, neither alone

D Even both

✓ **Verified Answer:** (C)** — `Both together are sufficient, neither alone` **Detailed Step-by-Step Solution & Logical Derivation:** Question: What is Riya's present age? - Statement I: R = 2S, S = B - 3. Three unknowns, two equations. Insufficient. - Statement II: (R + 5) + (S + 5) = 40 => R + S = 30. Two unknowns, one equation. Insufficient. - Both together: Substitute R = 2S into R + S = 30 => 2S + S = 30 => 3S = 30 => S = 10. Thus R = 20 years. Riya's present age is uniquely determined as 20. Both statements together are sufficient, neither alone.

Problem 5 • Infosys SE Question 5

MUST-SOLVE

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_01_page_01.png

Question Text:
In a certain code, MONDAY is written as NNOCBX. How is SUNDAY written in that code?

A TTOCBX

B TVOEBZ

C TTOCBZ

D RTOCBX

✓ **Verified Answer:** (A)** — `TTOCBX` **Detailed Step-by-Step Solution & Logical Derivation:** Coding Pattern: M (+1) -> N, O (-1) -> N, N (+1) -> O, D (-1) -> C, A (+1) -> B, Y (-1) -> X. The alternating shift pattern is [+1, -1, +1, -1, +1, -1]. Applying to SUNDAY: - S (+1) -> T - U (-1) -> T - N (+1) -> O - D (-1) -> C - A (+1) -> B - Y (-1) -> X Result: TTOCBX.

Problem 6 • Infosys SE Question 6

MUST-SOLVE

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_01_page_01.png

Question Text:
In a code, "sun rises early" = "ka mo ti", "early bird catches" = "ti ra po", and "bird sun light" = "ra ka du".
What is the code for "rises bird light"?

A mo ra du

B mo po du

C ka ra du

D mo ti du

✓ **Verified Answer:** (A)** — `mo ra du` **Detailed Step-by-Step Solution & Logical Derivation:** Word-code comparison: 1) 'sun rises early' = 'ka mo ti' 2) 'early bird catches' = 'ti ra po' => Common word 'early' = 'ti'. 3) 'bird sun light' = 'ra ka du' => Compare (1) & (3): common word 'sun' = 'ka'. From (1), remaining word 'rises' = 'mo'. Compare (2) & (3): common word 'bird' = 'ra'. From (3), remaining word 'light' = 'du'. Target: 'rises bird light' = 'mo ra du'.

Problem 7 • Infosys SE Question 7

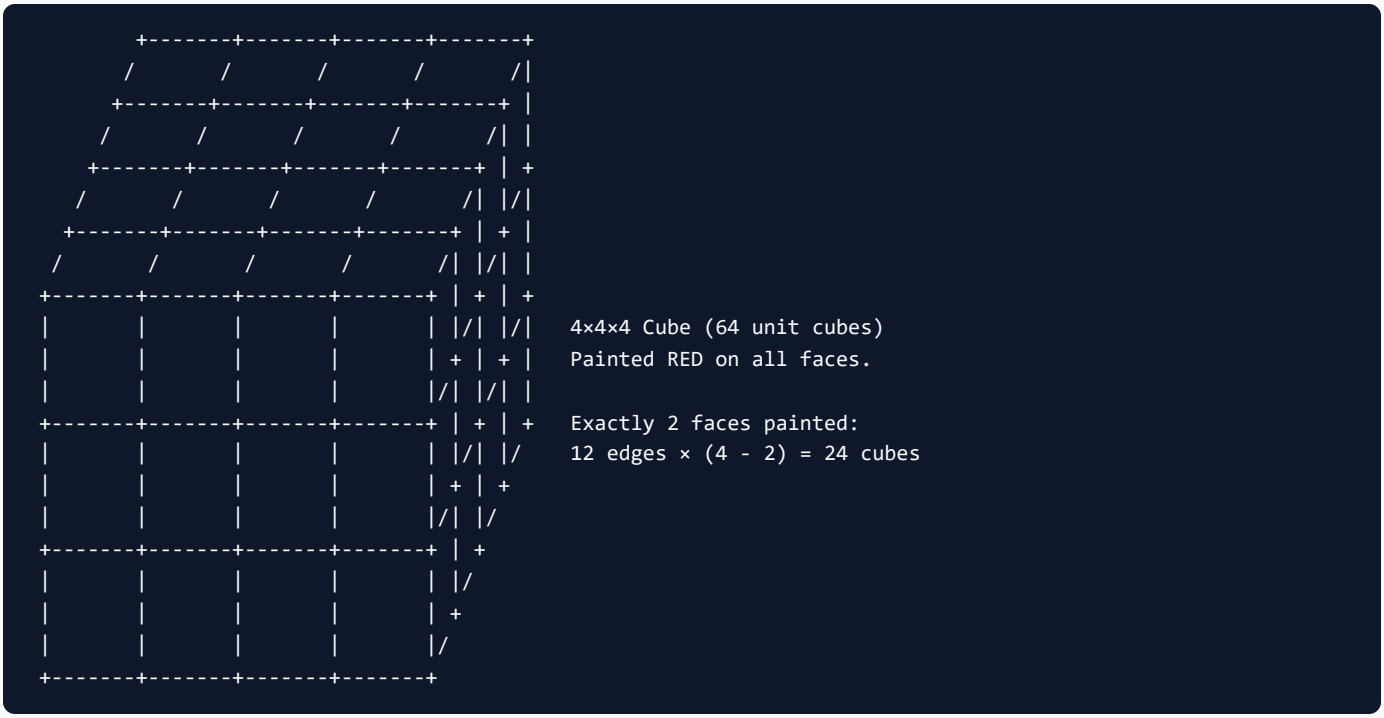
MUST-SOLVE

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

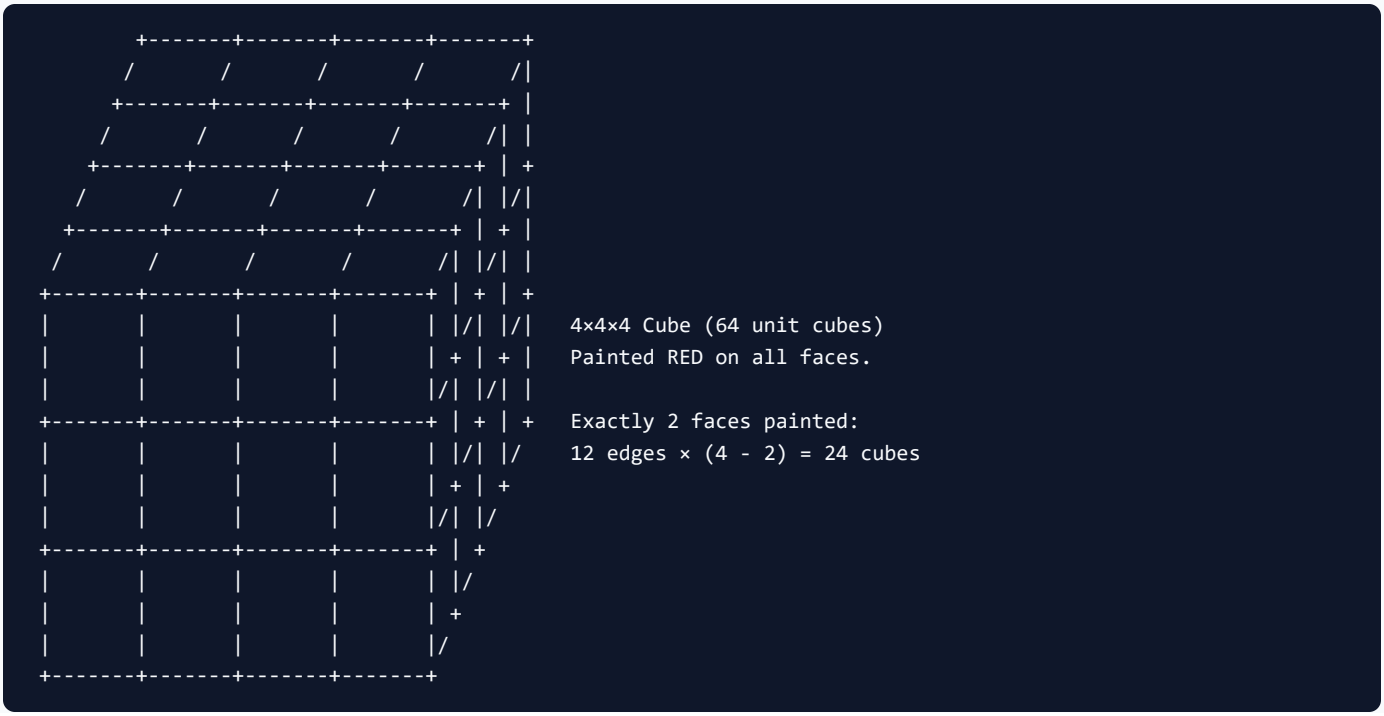
(Infosys System Engineer ITM Mock Paper 1 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_01_page_01.png

Question Text:
A solid cube painted red on all faces is cut into 64 identical small cubes (4 × 4 × 4). How many small cubes have exactly two faces painted?

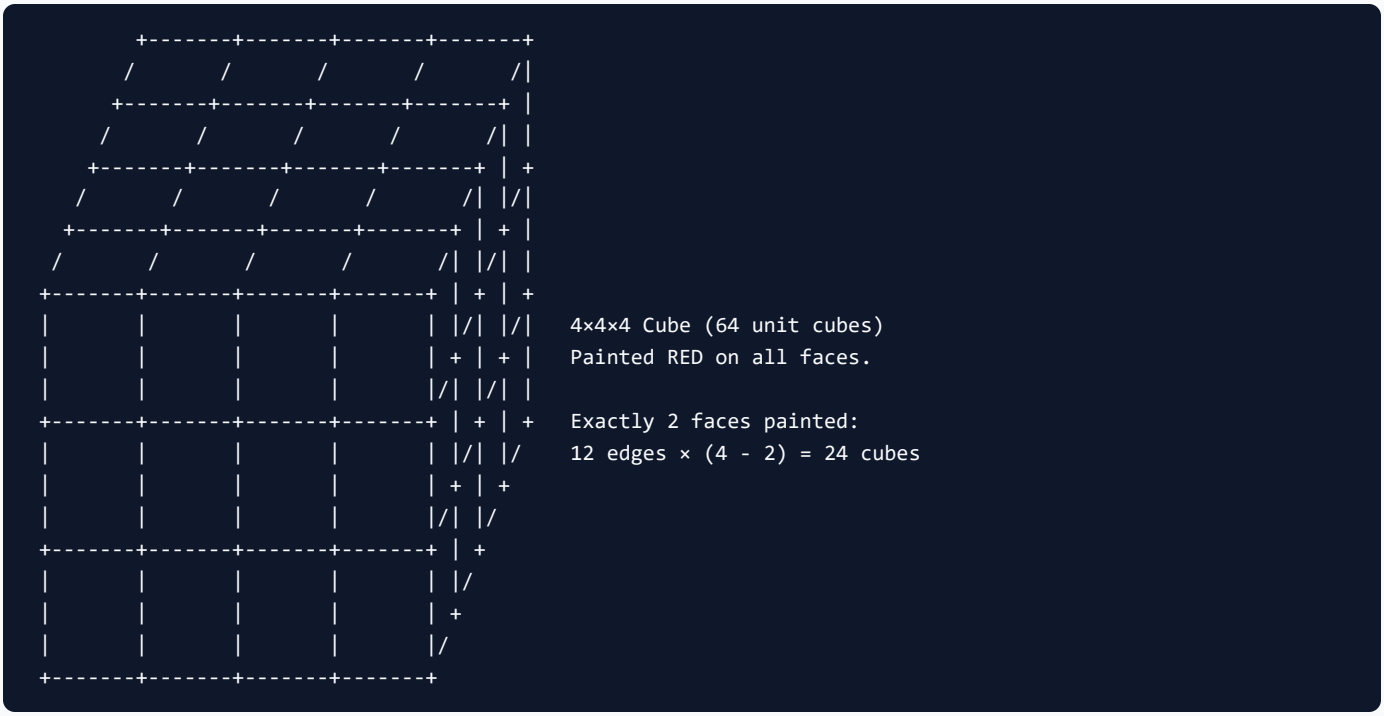
Visual Schematic / Logical Deduction Diagram:



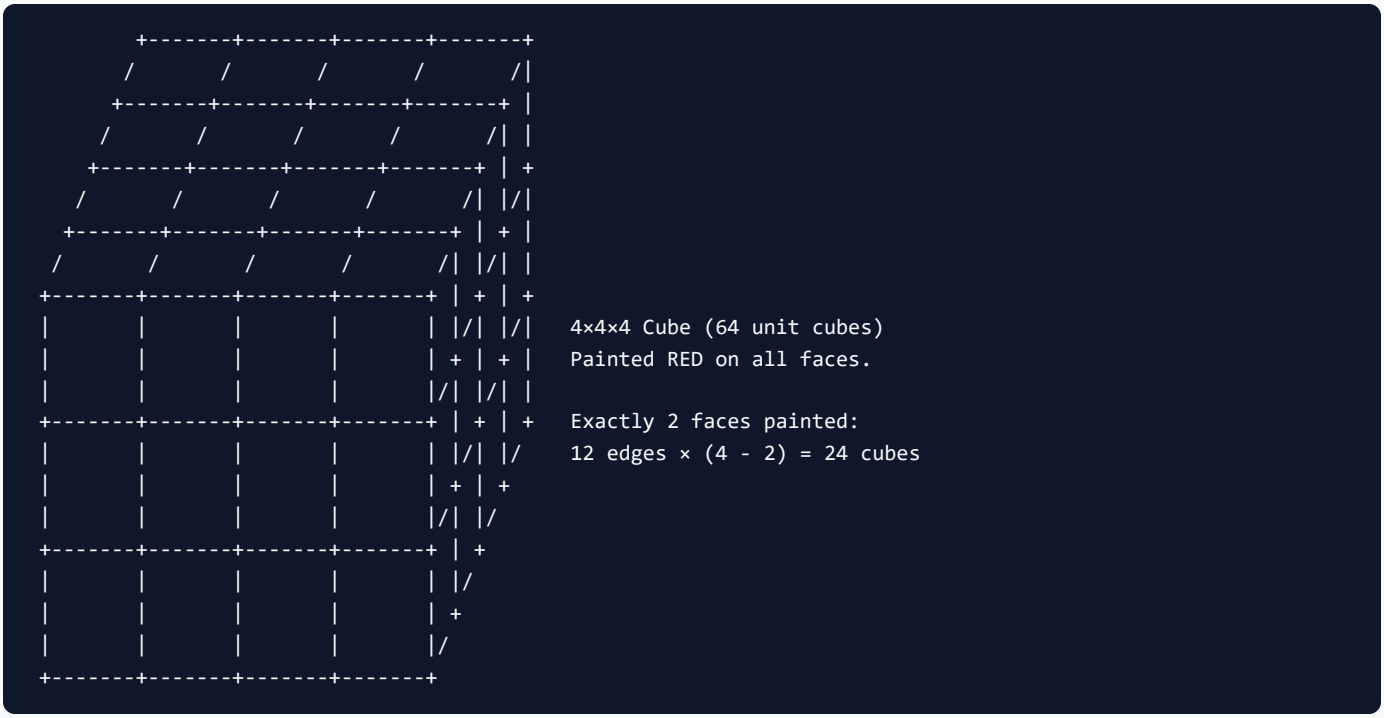
Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



A 12

B 24

C 36

D 8

Verified Answer: (B)** — `24` **Detailed Step-by-Step Solution & Logical Derivation:** For an n × n × n painted cube cut into unit cubes (here n = 4 since $4^3 = 64$): Cubes with exactly two faces painted lie along the 12 edges (excluding the 2 corner cubes on

✓ each edge): Number = 12 × (n - 2) = 12 × (4 - 2) = 12 × 2 = 24.

Problem 8 • Infosys SE Question 8

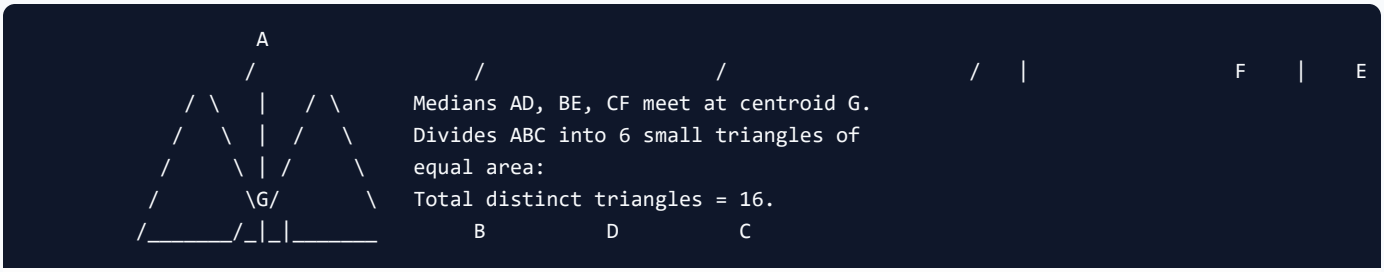
MUST-SOLVE

ID: Q8 | Source: Infosys PYQ | Level: Infosys SE Standard

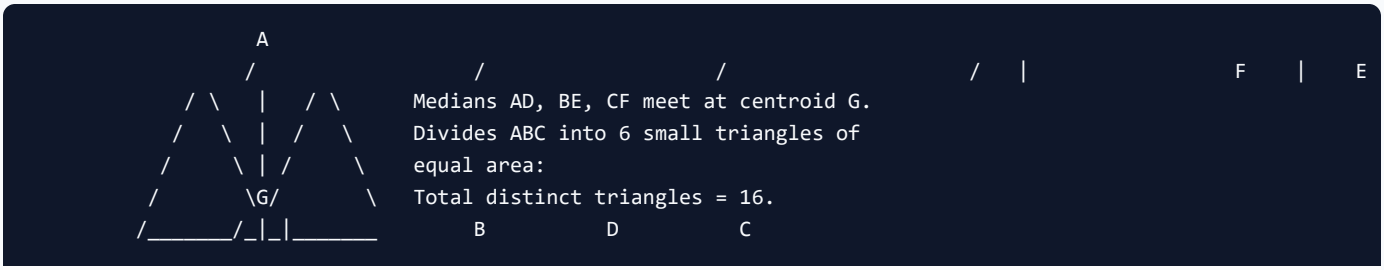
(Infosys System Engineer ITM Mock Paper 1 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
In triangle ABC, the medians AD, BE and CF are drawn and meet at G. How many triangles are there in the figure in total?

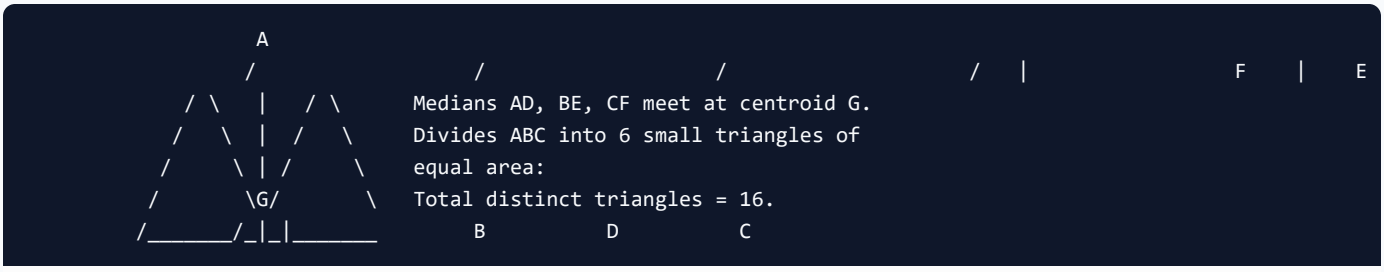
Visual Schematic / Logical Deduction Diagram:



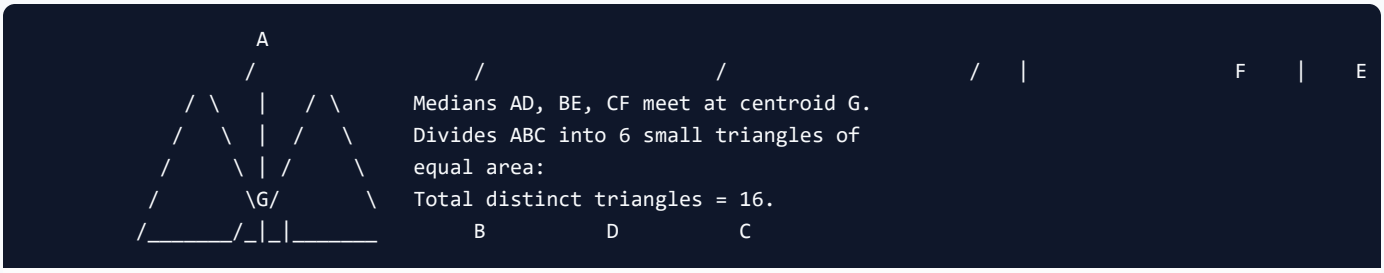
Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



A 12

B 14

C 16

D 18

✓ **Verified Answer:** (C)** — `16` **Detailed Step-by-Step Solution & Logical Derivation:** Standard median geometry counting: In triangle ABC with medians AD, BE, CF intersecting at centroid G: 1) 6 small elementary triangles formed around centroid G: AFG, BFG, BDG, CDG, CEG, AEG. 2) 3 triangles formed by pairing two adjacent elementary triangles opposite vertices: ABG, BCG, CAG. 3) 6 triangles formed by combining three elementary triangles (split by each median into halves): ABD, ACD, BCE, ABE, CAF, CBF. 4) 1 whole outer triangle: ABC. Total = 6 + 3 + 6 + 1 = 16 triangles.

Problem 9 • Infosys SE Question 9

MUST-SOLVE

ID: Q9 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
Pointing to a man, Meera said, "His mother is the only daughter of my mother." How is Meera related to the man?

Visual Schematic / Logical Deduction Diagram:

```
          [Grandmother]
               |
       [Mother (Meera)]
               |
          [Son (Man)]
"His mother is the only daughter of my mother."
=> Meera is the mother of the man.
```

Visual Schematic / Logical Deduction Diagram:

```
          [Grandmother]
               |
       [Mother (Meera)]
               |
          [Son (Man)]
"His mother is the only daughter of my mother."
=> Meera is the mother of the man.
```

Visual Schematic / Logical Deduction Diagram:

```
          [Grandmother]
               |
       [Mother (Meera)]
               |
          [Son (Man)]
"His mother is the only daughter of my mother."
=> Meera is the mother of the man.
```

Visual Schematic / Logical Deduction Diagram:

```
          [Grandmother]
               |
       [Mother (Meera)]
               |
          [Son (Man)]
"His mother is the only daughter of my mother."
=> Meera is the mother of the man.
```

A Mother

B Sister

C Aunt

D Grandmother

✓ **Verified Answer:** (A)** — `Mother` **Detailed Step-by-Step Solution & Logical Derivation:** Meera said: 'His mother is the only daughter of my mother.' The 'only daughter of my mother' when spoken by a woman (Meera) refers to Meera herself. Therefore, 'His mother is Meera.' Thus, Meera is the man's Mother.

Problem 10 • Infosys SE Question 10

MUST-SOLVE

`ID: Q10` `Source: Infosys PYQ` `Level: Infosys SE Standard`

(Infosys System Engineer ITM Mock Paper 1 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
If A + B means A is the mother of B, A − B means A is the brother of B, and A × B means A is the father of B, which expression shows that P is the maternal uncle of R?

A P + Q − R

B P − Q + R

C P × Q − R

D P − Q × R

✓ **Verified Answer:** (B)** — `P − Q + R` **Detailed Step-by-Step Solution & Logical Derivation:** P must be the maternal uncle of R (brother of R's mother). - Q + R means Q is the mother of R. - P - Q means P is the brother of Q. Combining them: P - Q + R means P is the brother of Q (who is the mother of R), so P is R's maternal uncle.

Problem 11 • Infosys SE Question 11

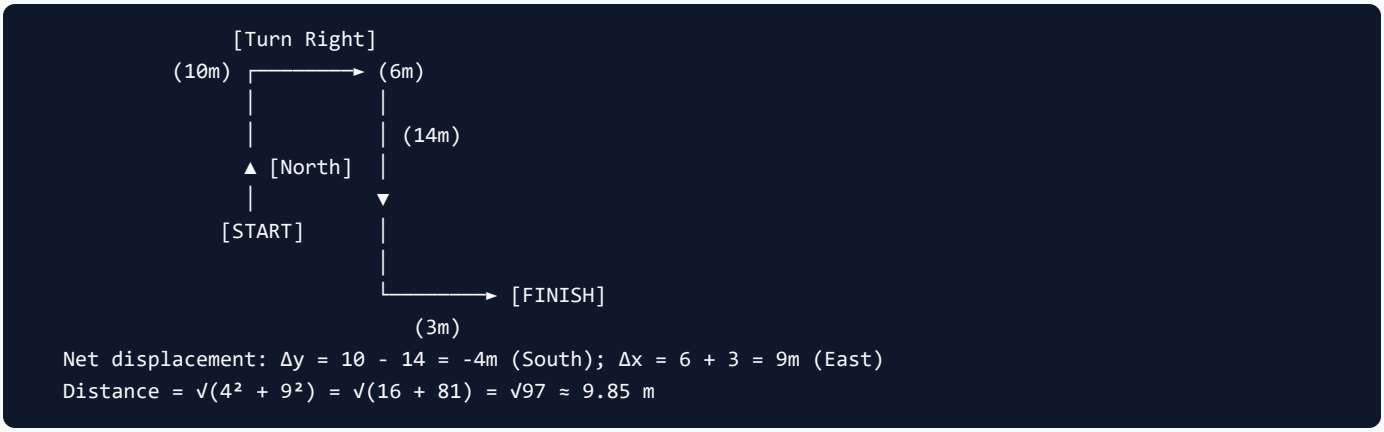
MUST-SOLVE

ID: Q11 | Source: Infosys PYQ | Level: Infosys SE Standard

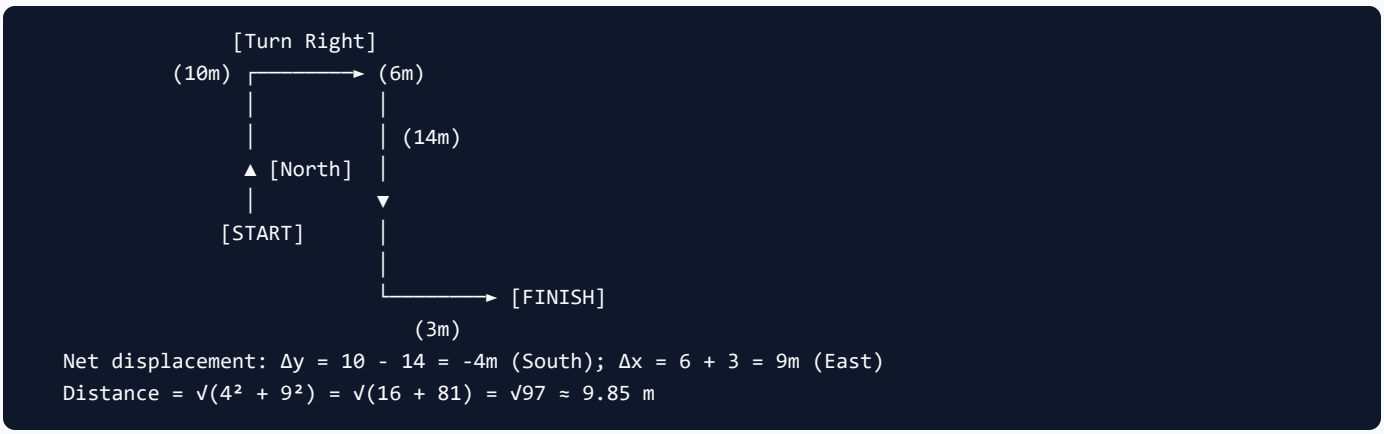
(Infosys System Engineer ITM Mock Paper 1 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
Ravi walks 10 m north, turns right and walks 6 m, turns right and walks 14 m, turns left and walks 3 m, then turns left and walks 4 m. Where is he now relative to the starting point?

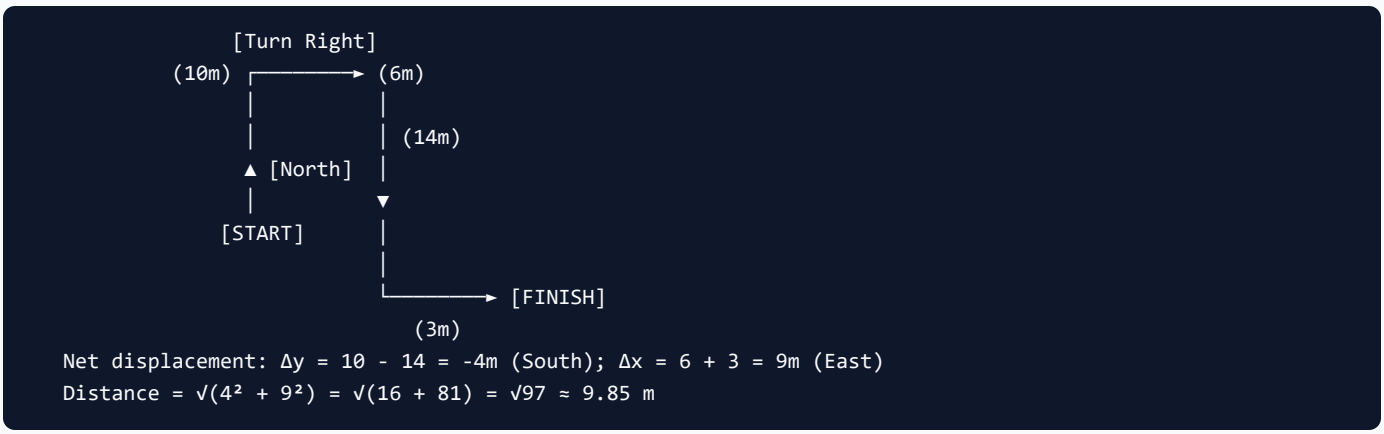
Visual Schematic / Logical Deduction Diagram:



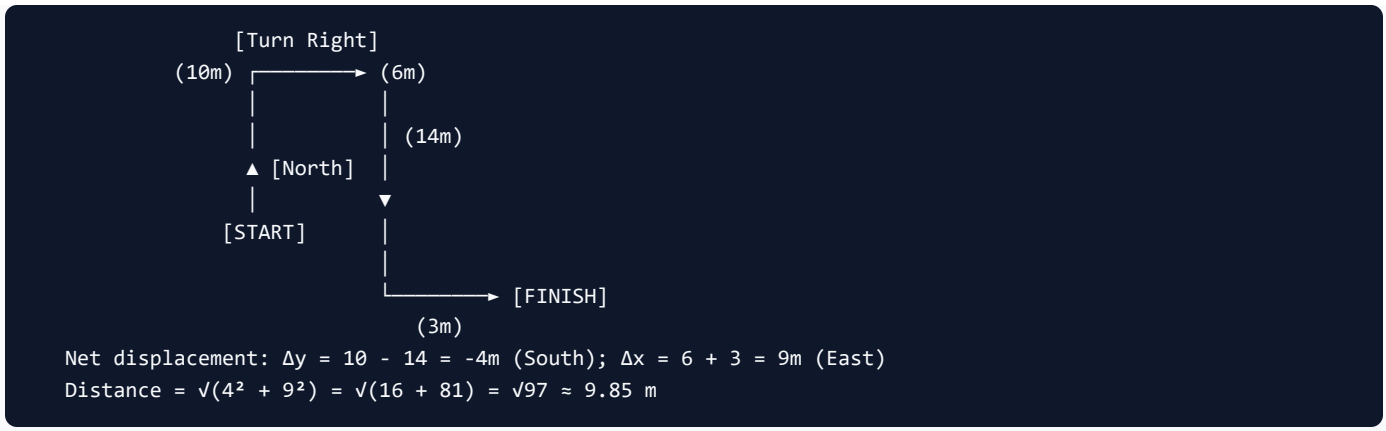
Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Net displacement: Δy = 10 - 14 = -4m (South); Δx = 6 + 3 = 9m (East)
Distance = √(4² + 9²) = √(16 + 81) = √97 ≈ 9.85 m

A 9 m East

B 9 m West

C 11 m South-East

D 5 m North

✓ **Verified Answer:** (A)** — `9 m East` **Detailed Step-by-Step Solution & Logical Derivation:** Step-by-step vector tracking from origin (0, 0): 1) Walks 10 m North -> (0, 10) 2) Turns right (East) and walks 6 m -> (6, 10) 3) Turns right (South) and walks 14 m -> (6, 10 - 14) = (6, -4) 4) Turns left (East) and walks 3 m -> (6 + 3, -4) = (9, -4) 5) Turns left (North) and walks 4 m -> (9, -4 + 4) = (9, 0) Final position is (9, 0), which is exactly 9 m East of the starting point.

Problem 12 • Infosys SE Question 12

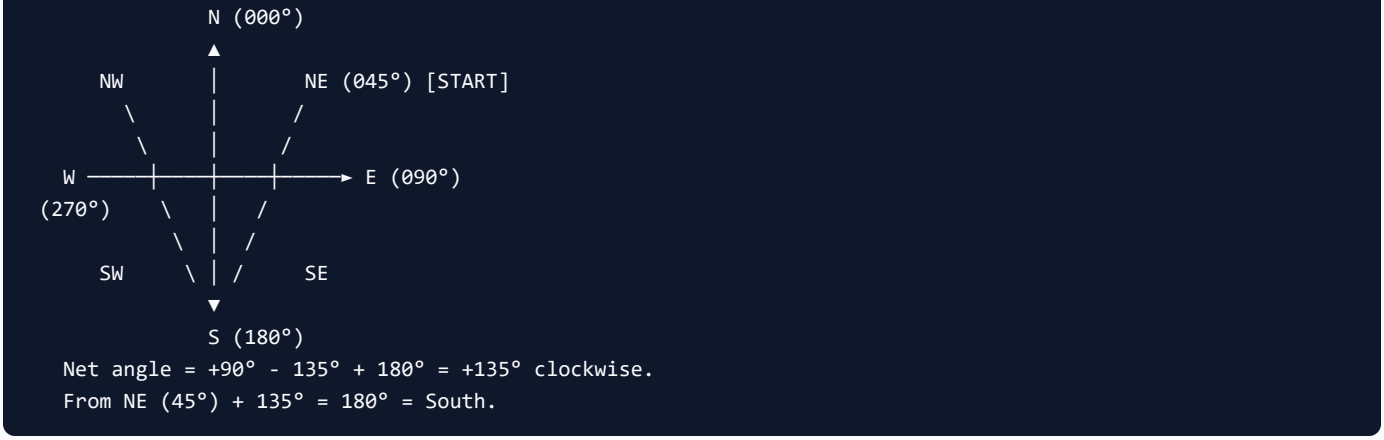
MUST-SOLVE

ID: Q12 | Source: Infosys PYQ | Level: Infosys SE Standard

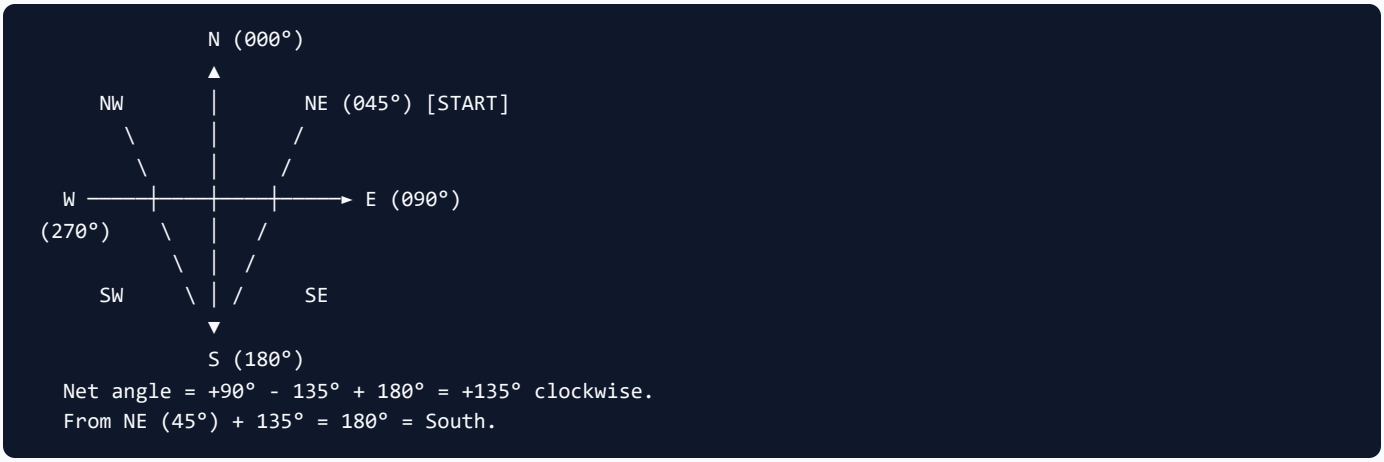
(Infosys System Engineer ITM Mock Paper 1 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
A person facing North-East turns 90° clockwise, then 135° anticlockwise, then 180° clockwise. Which direction is he facing now?

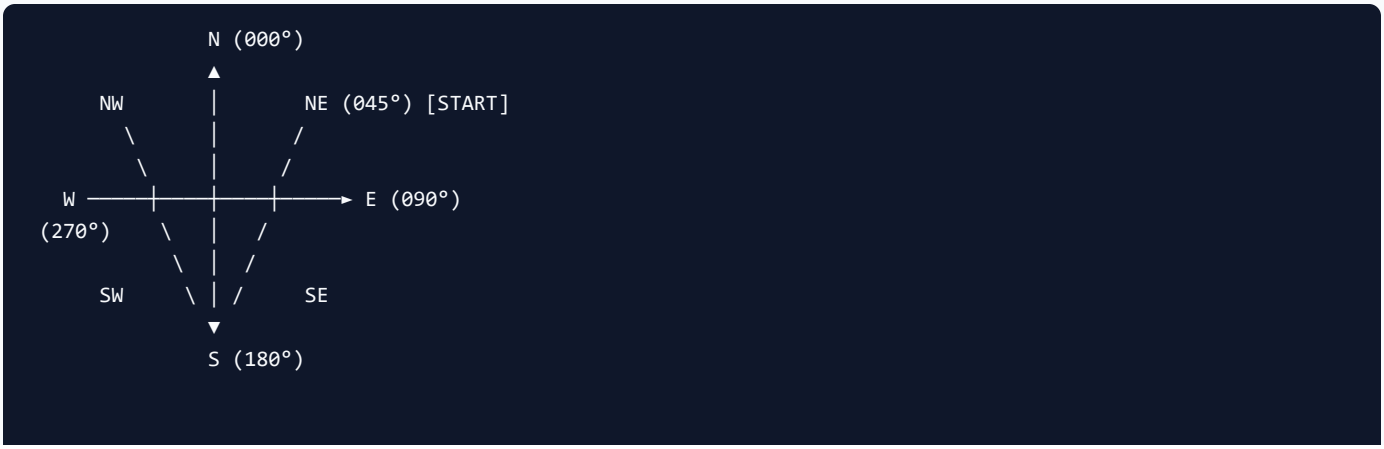
Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:

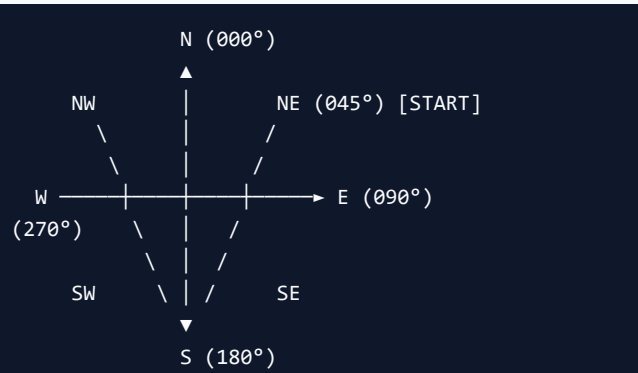


Visual Schematic / Logical Deduction Diagram:



```
Net angle = +90° - 135° + 180° = +135° clockwise.
From NE (45°) + 135° = 180° = South.
```

Visual Schematic / Logical Deduction Diagram:




```
Net angle = +90° - 135° + 180° = +135° clockwise.
From NE (45°) + 135° = 180° = South.
```

A North

B South

C South-West

D East

✓ **Verified Answer:** (B)** — `South` **Detailed Step-by-Step Solution & Logical Derivation:** Initial direction: North-East (45° clockwise from North). 1) +90° clockwise: 45° + 90° = 135° (South-East). 2) -135° anticlockwise: 135° - 135° = 0° (North). 3) +180° clockwise: 0° + 180° = 180° (South). Final direction is South.

Problem 13 • Infosys SE Question 13

MUST-SOLVE

ID: Q13 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
Find the next term: 3, 7, 16, 35, 74, ?

A 143

B 153

C 149

D 158

✓ **Verified Answer:** (B)** — `153` **Detailed Step-by-Step Solution & Logical Derivation:** Series: 3, 7, 16, 35, 74, ? Pattern: ×2 + k where k increments by 1 at each step: - 3 × 2 + 1 = 7 - 7 × 2 + 2 = 16 - 16 × 2 + 3 = 35 - 35 × 2 + 4 = 74 - Next term = 74 × 2 + 5 = 148 + 5 = 153.

Problem 14 • Infosys SE Question 14

MUST-SOLVE

ID: Q14 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q14)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
Find the next term: 4, 9, 25, 49, 121, 169, ?

A 289

B 259

C 361

D 323

✓ **Verified Answer:** (A)** — `289` **Detailed Step-by-Step Solution & Logical Derivation:** Series: 4, 9, 25, 49, 121, 169, ? Notice the base numbers: - $2^2 = 4$ - $3^2 = 9$ - $5^2 = 25$ - $7^2 = 49$ - $11^2 = 121$ - $13^2 = 169$ These are the squares of consecutive prime numbers (2, 3, 5, 7, 11, 13). The next prime number is 17. Next term = $17^2 = 289$.

Problem 15 • Infosys SE Question 15

MUST-SOLVE

ID: Q15 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q15)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
Find the next letter: B, E, I, N, T, ?

A A

B Z

C B

D Y

✓ **Verified Answer:** (A)** — `A` **Detailed Step-by-Step Solution & Logical Derivation:** Letter positions in alphabet: B(2), E(5), I(9), N(14), T(20), ? Differences: - 5 - 2 = +3 - 9 - 5 = +4 - 14 - 9 = +5 - 20 - 14 = +6 Next difference is +7: 20 + 7 = 27. In circular alphabet (mod 26): 27 ≡ 1 = A.

Problem 16 • Infosys SE Question 16

MUST-SOLVE

ID: Q16 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_02_page_01.png

Question Text:
Statements: All servers are machines. Some machines are laptops. No laptop is portable.
Conclusions: I. Some machines are not portable. II. Some servers are laptops. III. No server is portable.

A Only I follows

B I and III follow

C I and II follow

D None follows

✓ **Verified Answer:** (A)** — `Only I follows` **Detailed Step-by-Step Solution & Logical Derivation:** Some machines are laptops, and no laptop is portable, so those machines are not portable — I follows. Nothing links servers specifically to laptops or to portability, so II and III don't follow.

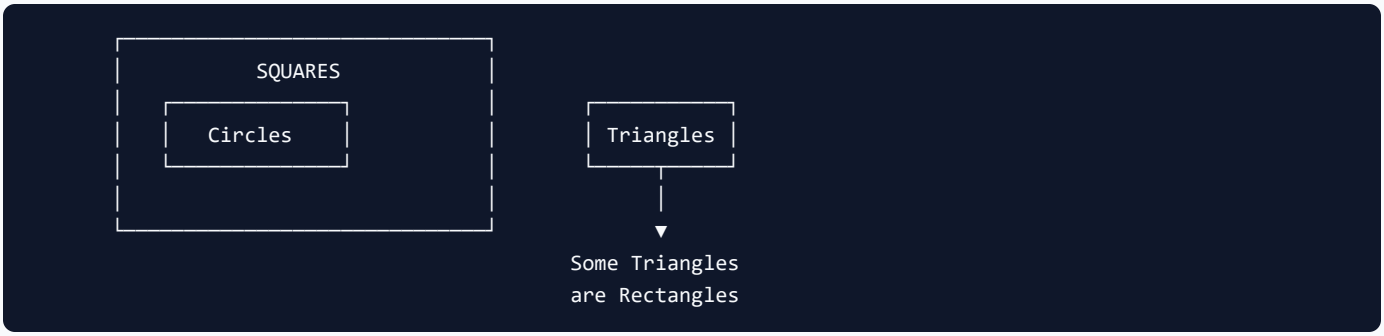
Problem 17 • Infosys SE Question 17 MUST-SOLVE

ID: Q17 | Source: Infosys PYQ | Level: Infosys SE Standard

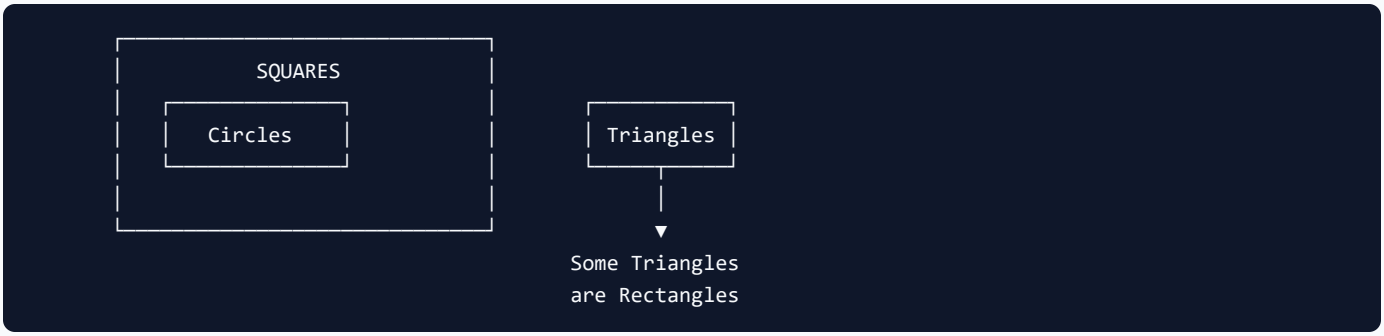
(Infosys System Engineer ITM Mock Paper 2 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_02_page_01.png

Question Text:
Statements: All circles are squares. No square is a triangle. Some triangles are rectangles.
Conclusions: I. No circle is a triangle. II. Some rectangles are not squares. III. Some squares are not triangles.

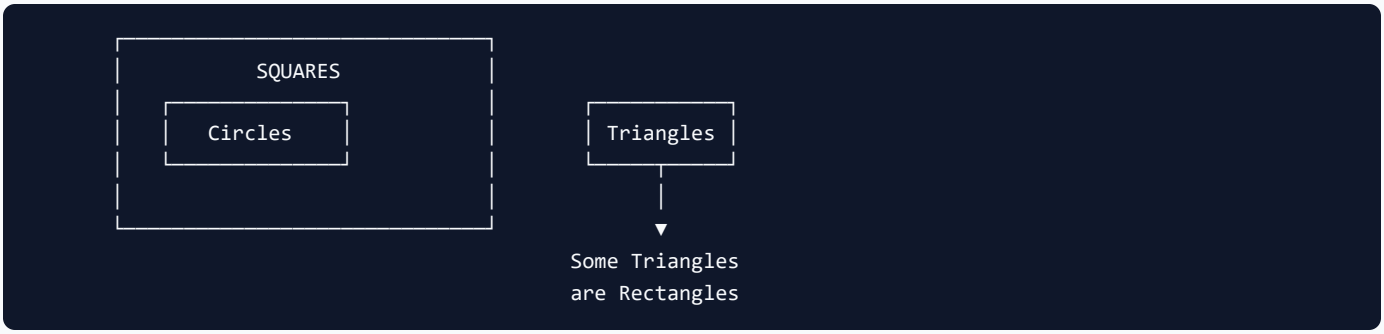
Visual Schematic / Logical Deduction Diagram:



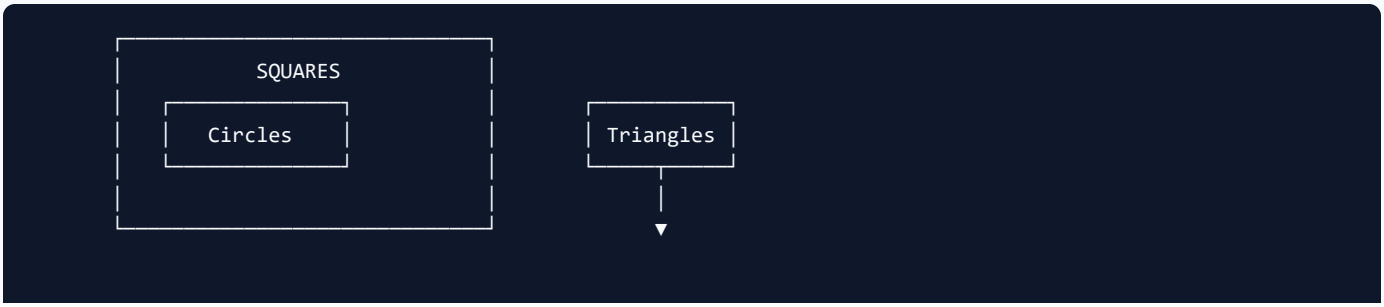
Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



Visual Schematic / Logical Deduction Diagram:



```
                                        Some Triangles
                                        are Rectangles
```

A I and III only

B I only

C I, II and III

D II and III only

✓ **Verified Answer:** (C)** — `I, II and III` **Detailed Step-by-Step Solution & Logical Derivation:** Circles squares, and squares have no overlap with triangles, so circles have none either (I). Some ⊆ triangles are rectangles, and since no triangle is a square, those rectangles are not squares (II). Since no square is a triangle, it's also true that some squares are not triangles (III).

Problem 18 • Infosys SE Question 18

MUST-SOLVE

`ID: Q18` `Source: Infosys PYQ` `Level: Infosys SE Standard`

(Infosys System Engineer ITM Mock Paper 2 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_02_page_01.png

Question Text:
Data Sufficiency. Is the integer n divisible by 6?
I. n is divisible by 4. II. n is divisible by 9.

A I alone is sufficient

B II alone is sufficient

C Both together are sufficient, neither alone

D Even both

✓ **Verified Answer:** (C)** — `Both together are sufficient, neither alone` **Detailed Step-by-Step Solution & Logical Derivation:** I alone fails (n = 4 is not divisible by 6). II alone fails (n = 9 is not divisible by 6). Together, a number divisible by both 4 and 9 is divisible by their LCM, 36, which is always divisible by 6.

Problem 19 • Infosys SE Question 19

MUST-SOLVE

ID: Q19 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_02_page_01.png

Question Text:
Data Sufficiency. What is the cost price of an article?
I. It was sold at a profit of 20%. II. If sold at ₹150 more, the profit would have been 32%.

A I alone is sufficient

B II alone is sufficient

C Both together are sufficient, neither alone

D Even both

✓ **Verified Answer:** (B)** — `II alone is sufficient` **Detailed Step-by-Step Solution & Logical Derivation:** II alone: 12% of CP = ₹150, so CP = ₹1,250 — sufficient. I alone gives only a percentage, not an absolute value.

Problem 20 • Infosys SE Question 20

MUST-SOLVE

ID: Q20 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_02_page_01.png

Question Text:
In a code, every letter is shifted 3 places forward in the alphabet (wrapping Z→C), so FLOWER is written as IORZHU. Using the same code, how is GARDEN written?

A JDUGHQ

B JDUFHQ

C JCUGHQ

D JDVGHQ

✓ **Verified Answer:** (A)** — `JDUGHQ` **Detailed Step-by-Step Solution & Logical Derivation:** Applying the same +3 shift: G→J, A→D, R→U, D→G, E→H, N→Q, giving JDUGHQ.

Problem 21 • Infosys SE Question 21

MUST-SOLVE

ID: Q21 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_02_page_01.png

Question Text:
In a code, "data flows fast" = "zu ka mo", "fast network grows" = "ka ta ri", "network data secure" = "zu ta ve". What is the code for "grows secure data"?

A ri ve zu

B ta ve zu

C ri ve ka

D ka ve zu

✓ **Verified Answer:** (A)** — `ri ve zu` **Detailed Step-by-Step Solution & Logical Derivation:** "fast" is common to statements 1 and 2 → ka. "data" is common to 1 and 3 → zu. "network" is common to 2 and 3 → ta. By elimination: flows = mo, grows = ri, secure = ve. So "grows secure data" = ri ve zu.

Problem 22 • Infosys SE Question 22

MUST-SOLVE

ID: Q22 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_02_page_01.png

Question Text:
A cube is painted red on one pair of opposite faces, blue on another pair, and green on the last pair. It is then cut into 27 identical small cubes. How many small cubes have exactly one face painted?

Visual Schematic / Logical Deduction Diagram:

```
        Top/Bottom: Green
        Front/Back: Red
        Left/Right: Blue
        4×4×4 Cube (64 unit cubes)
At least two different colors = (Corners: 8) + (Edges: 24) = 32 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
        Top/Bottom: Green
        Front/Back: Red
        Left/Right: Blue
        4×4×4 Cube (64 unit cubes)
At least two different colors = (Corners: 8) + (Edges: 24) = 32 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
        Top/Bottom: Green
        Front/Back: Red
        Left/Right: Blue
        4×4×4 Cube (64 unit cubes)
At least two different colors = (Corners: 8) + (Edges: 24) = 32 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
        Top/Bottom: Green
        Front/Back: Red
        Left/Right: Blue
        4×4×4 Cube (64 unit cubes)
At least two different colors = (Corners: 8) + (Edges: 24) = 32 cubes.
```

A 6

B 8

C 9

D 12

✓ **Verified Answer:** (A)** — `6` **Detailed Step-by-Step Solution & Logical Derivation:** In a 3×3×3 cube, only the centre cube of each face touches just one painted face — 1 per face × 6 faces = 6, regardless of the colour scheme.

Problem 23 • Infosys SE Question 23

MUST-SOLVE

ID: Q23 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_02_page_01.png

Question Text:
A rectangle is divided into a 3 × 3 grid of smaller cells by 2 evenly spaced vertical lines and 2 evenly spaced horizontal lines drawn inside it. How many rectangles (of any size) are there in the whole figure?

A 30

B 36

C 27

D 24

✓ **Verified Answer:** (B)** — `36` **Detailed Step-by-Step Solution & Logical Derivation:** With 2 interior + 2 boundary lines in each direction, there are 4 vertical and 4 horizontal lines. Rectangles = C(4,2) × C(4,2) = 6 × 6 = 36.

Problem 24 • Infosys SE Question 24

MUST-SOLVE

ID: Q24 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
Pointing to a photograph, a man says, "She is the daughter of my grandfather's only son." How is the woman in the photograph related to the man?

Visual Schematic / Logical Deduction Diagram:

```
            [Grandfather]
                 |
            [Only Son] (Father)
                 |
          ┌──────┴──────┐
          |             |
      [Man]         [Daughter] (Sister)
=> "She is the daughter of my grandfather's only son" = Sister.
```

Visual Schematic / Logical Deduction Diagram:

```
            [Grandfather]
                 |
            [Only Son] (Father)
                 |
          ┌──────┴──────┐
          |             |
      [Man]         [Daughter] (Sister)
=> "She is the daughter of my grandfather's only son" = Sister.
```

Visual Schematic / Logical Deduction Diagram:

```
            [Grandfather]
                 |
            [Only Son] (Father)
                 |
          ┌──────┴──────┐
          |             |
      [Man]         [Daughter] (Sister)
=> "She is the daughter of my grandfather's only son" = Sister.
```

Visual Schematic / Logical Deduction Diagram:

```
            [Grandfather]
                 |
            [Only Son] (Father)
                 |
          ┌──────┴──────┐
          |             |
      [Man]         [Daughter] (Sister)
=> "She is the daughter of my grandfather's only son" = Sister.
```

A Sister

B Daughter

C Niece

D Mother

✓ **Verified Answer:** (A)** — `Sister` **Detailed Step-by-Step Solution & Logical Derivation:** The grandfather's only son is the man's father. The daughter of the man's father is his sister.

Problem 25 • Infosys SE Question 25

MUST-SOLVE

`ID: Q25` `Source: Infosys PYQ` `Level: Infosys SE Standard`

(Infosys System Engineer ITM Mock Paper 2 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
If A ÷ B means A is the daughter of B, A + B means A is the son of B, and A − B means A is the wife of B, which expression correctly shows that M is the granddaughter of N?

A M ÷ P + N

B M + P ÷ N

C M − P ÷ N

D M ÷ P − N

✓ **Verified Answer:** (A)** — `M ÷ P + N` **Detailed Step-by-Step Solution & Logical Derivation:** M ÷ P: M is daughter of P. P + N: P is son of N. So M is P's daughter, and P is N's son — M is N's granddaughter.

Problem 26 • Infosys SE Question 26 MUST-SOLVE

ID: Q26 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
Priya starts at a point, faces south and walks 8 m. She turns left and walks 5 m, turns left again and walks 8 m, then turns right and walks 3 m. How far is she from the starting point, and in which direction?

Visual Schematic / Logical Deduction Diagram:

```
          [START]
             |
             ▼ (8m South)
             |
    (5m) ◄──┘ [Turn Left = East]
     ▲
     | (12m North)
     |
  [FINISH]
Net displacement: Δy = 12 - 8 = 4m North; Δx = 5m East
Distance = √(4² + 5²) = √(16 + 25) = √41 m
```

Visual Schematic / Logical Deduction Diagram:

```
          [START]
             |
             ▼ (8m South)
             |
    (5m) ◄──┘ [Turn Left = East]
     ▲
     | (12m North)
     |
  [FINISH]
Net displacement: Δy = 12 - 8 = 4m North; Δx = 5m East
Distance = √(4² + 5²) = √(16 + 25) = √41 m
```

Visual Schematic / Logical Deduction Diagram:

```
          [START]
             |
             ▼ (8m South)
             |
    (5m) ◄──┘ [Turn Left = East]
     ▲
     | (12m North)
     |
  [FINISH]
Net displacement: Δy = 12 - 8 = 4m North; Δx = 5m East
Distance = √(4² + 5²) = √(16 + 25) = √41 m
```

Visual Schematic / Logical Deduction Diagram:

```
          [START]
             |
```

```
              ▼ (8m South)
              |
      (5m) ◄──┘ [Turn Left = East]
       ▲
       | (12m North)
       |
    [FINISH]
Net displacement: Δy = 12 - 8 = 4m North; Δx = 5m East
Distance = √(4² + 5²) = √(16 + 25) = √41 m
```

A 8 m East

B 8 m West

C 2 m East

D 2 m West

✓ **Verified Answer:** (A)** — `8 m East` **Detailed Step-by-Step Solution & Logical Derivation:** Tracking positions from (0,0): south 8 → (0,−8); left turn faces east, walk 5 → (5,−8); left turn faces north, walk 8 → (5,0); right turn faces east, walk 3 → (8,0). Net displacement is 8 m East.

Problem 27 • Infosys SE Question 27

CORE SE

ID: Q27 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A man facing South turns 45° anticlockwise, then 90° clockwise, then 225° anticlockwise. Which direction is he facing now?

A North

B South-East

C South-West

D West

✓ **Verified Answer:** (A)** — `North` **Detailed Step-by-Step Solution & Logical Derivation:** Using bearings (clockwise from North): South = 180°. −45° (anticlockwise) = 135°; +90° (clockwise) = 225°; −225° (anticlockwise) = 0°, which is North.

Problem 28 • Infosys SE Question 28

CORE SE

ID: Q28 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
Find the next term: 2, 6, 15, 31, 56, ?

A 92

B 88

C 90

D 84

✓ **Verified Answer:** (A)** — `92` **Detailed Step-by-Step Solution & Logical Derivation:** Differences are 4, 9, 16, 25 (perfect squares), so the next difference is 36: 56 + 36 = 92.

Problem 29 • Infosys SE Question 29

CORE SE

ID: Q29 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q14)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
Find the next term: 5, 11, 23, 47, 95, ?

A 189

B 191

C 193

D 190

✓ **Verified Answer:** (B)** — `191` **Detailed Step-by-Step Solution & Logical Derivation:** Each term = previous × 2 + 1: 95 × 2 + 1 = 191.

Problem 30 • Infosys SE Question 30

CORE SE

ID: Q30 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q15)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
Find the next letter-pair: CE, FI, IM, LQ, ?

A OT

B OU

C NT

D PU

✓ **Verified Answer:** (B)** — `OU` **Detailed Step-by-Step Solution & Logical Derivation:** First letters C, F, I, L rise by 3 each time → next is O. Second letters E, I, M, Q rise by 4 each time → next is U. Answer: OU.

Problem 31 • Infosys SE Question 31

CORE SE

ID: Q31 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_03_page_01.png

Question Text:
Statements: All P are Q. All Q are R. Some R are S. No S is T.
Conclusions: I. All P are R. II. Some Q are S. III. No P is T. IV. Some R are not T.

A Only I follows

B I and IV follow

C I, II and IV follow

D All follow

✓ **Verified Answer:** (B)** — `I and IV follow` **Detailed Step-by-Step Solution & Logical Derivation:** Since P⊆Q⊆R, "All P are R" (I) follows directly. Some R are S, and no S is T, so those particular R- elements are not T — "Some R are not T" (IV) follows. Neither II nor III can be derived, since we don't know how P or Q specifically relates to the R-elements that happen to be S.

Problem 32 • Infosys SE Question 32

CORE SE

ID: Q32 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_03_page_01.png

Question Text:
Statements: All coins are metals. No metal is wood. Some woods are furniture.
Conclusions: I. No coin is wood. II. Some furniture is not metal.

A Only I follows

B Only II follows

C Both I and II follow

D Neither follows

✓ **Verified Answer:** (C)** — `Both I and II follow` **Detailed Step-by-Step Solution & Logical Derivation:** Coins⊆metals, and metals∩wood=∅, so coins∩wood=∅ — I follows. Some woods are furniture, and since no metal is wood, those wood-furniture items are not metal — II follows.

Problem 33 • Infosys SE Question 33

CORE SE

ID: Q33 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_03_page_01.png

Question Text:
Data Sufficiency. What is the two-digit number N?
I. The sum of its digits is 9.
II. The number obtained by reversing its digits is 27 less than N.

A Statement I alone is sufficient, but II alone is not

B Statement II alone is sufficient, but I alone is not

C Both statements together are sufficient, but neither alone is

D Each statement alone is sufficient

✓ **Verified Answer:** (C)** — `Both statements together are sufficient, but neither alone is` **Detailed Step-by-Step Solution & Logical Derivation:** II alone: N − reversed = 9(tens digit − units digit) = 27, so tens − units = 3 — multiple numbers fit (41, 52, 63, 74, 85, 96), insufficient alone. I alone: digit sum = 9 also fits multiple numbers, insufficient alone. Together: tens − units = 3 and tens + units = 9 gives tens = 6, units = 3, so N = 63 — unique.

Problem 34 • Infosys SE Question 34

CORE SE

ID: Q34 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_03_page_01.png

Question Text:
Data Sufficiency. Is the integer x even?
I. x/2 is an integer.
II. x^2 is even.

A Statement I alone is sufficient, but II alone is not

B Statement II alone is sufficient, but I alone is not

C Both statements together are sufficient, but neither alone is

D Each statement alone is sufficient

✓ **Verified Answer:** (D)** — `Each statement alone is sufficient (e) Even both statements together are not sufficient` **Detailed Step-by-Step Solution & Logical Derivation:** I alone: x/2 being an integer directly means x is even — sufficient. II alone: x^2 even is only possible if x itself is even (an odd number squared is always odd) — sufficient. Each statement alone is sufficient.

Problem 35 • Infosys SE Question 35

CORE SE

ID: Q35 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_03_page_01.png

Question Text:
In a certain code, TRAIN is written as 23145, GAME is written as 6178, and MEAT is written as 7812.
What is the code for GATE?

A 6128

B 6218

C 6182

D 8126

✓ **Verified Answer:** (A)** — `6128` **Detailed Step-by-Step Solution & Logical Derivation:** From TRAIN=23145: T=2,R=3,A=1,I=4,N=5. From GAME=6178: G=6,A=1,M=7,E=8. From MEAT=7812: confirms M=7,E=8,A=1,T=2. So GATE = G(6)A(1)T(2)E(8) = 6128.

Problem 36 • Infosys SE Question 36

CORE SE

ID: Q36 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
In a code: "bright light shines" = "mo ta pu", "shines through night" = "pu ri ke", "night bright stars" = "ke mo lo", "stars through galaxy" = "lo ri zu". What is the code for "night light galaxy"?

A ke ta zu

B ke ta ri

C mo ta zu

D ke lo zu

✓ **Verified Answer:** (A)** — `ke ta zu` **Detailed Step-by-Step Solution & Logical Derivation:** Comparing statement 1 & 2, the common word "shines" gets the common code "pu". Comparing 2 & 3, "night" gets the common code "ke". Comparing 1 & 3, "bright" gets the common code "mo". This leaves light=ta, stars=lo, through=ri, galaxy=zu (verified consistent with statement 4). So "night light galaxy" = ke ta zu.

Problem 37 • Infosys SE Question 37

CORE SE

ID: Q37 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
A cuboid measuring 5 × 4 × 3 units is painted on all its outer surfaces and then cut into unit cubes. How many unit cubes have exactly 2 painted faces?

Visual Schematic / Logical Deduction Diagram:

```
5 × 4 × 3 Cuboid (60 unit cubes)
Total Painted Surface Cubes:
No face painted = (5-2) × (4-2) × (3-2) = 3 × 2 × 1 = 6 cubes.
Painted cubes = 60 - 6 = 54 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
5 × 4 × 3 Cuboid (60 unit cubes)
Total Painted Surface Cubes:
No face painted = (5-2) × (4-2) × (3-2) = 3 × 2 × 1 = 6 cubes.
Painted cubes = 60 - 6 = 54 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
5 × 4 × 3 Cuboid (60 unit cubes)
Total Painted Surface Cubes:
No face painted = (5-2) × (4-2) × (3-2) = 3 × 2 × 1 = 6 cubes.
Painted cubes = 60 - 6 = 54 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
5 × 4 × 3 Cuboid (60 unit cubes)
Total Painted Surface Cubes:
No face painted = (5-2) × (4-2) × (3-2) = 3 × 2 × 1 = 6 cubes.
Painted cubes = 60 - 6 = 54 cubes.
```

- **A** 24
- **B** 22
- **C** 20
- **D** 26

✓ **Verified Answer:** (A)** — `24` **Detailed Step-by-Step Solution & Logical Derivation:** For a cuboid a×b×c, the number of unit cubes with exactly 2 painted faces = 4[(a−2)+(b−2)+(c−2)]. With a=5,b=4,c=3: 4[3+2+1] = 4×6 = 24.

Problem 38 • Infosys SE Question 38

CORE SE

ID: Q38 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
How many triangles are there in total in a complete five-pointed star (pentagram) figure?

A 5

B 10

C 35

D 15

✓ **Verified Answer:** (C)** — `35` **Detailed Step-by-Step Solution & Logical Derivation:** A complete five-pointed star (pentagram) figure contains 35 triangles in total (5 small tip triangles, 5 medium, 5 larger, and combinations up to the whole star) — a standard, well-documented result for this figure.

Problem 39 • Infosys SE Question 39

CORE SE

ID: Q39 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
Introducing a man, a woman says, "His wife is the only daughter-in-law of my father." How is the woman related to the man's wife?

A Sister-in-law

B Sister

C Mother

D Aunt

✓ **Verified Answer:** (A)** — `Sister-in-law` **Detailed Step-by-Step Solution & Logical Derivation:** The man's wife is the only daughter-in-law of the woman's father, meaning the woman's father has only one son — the man himself. So the man is the woman's brother, making the woman the sister-in-law of the man's wife.

Problem 40 • Infosys SE Question 40

CORE SE

ID: Q40 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
If A % B means A is the mother of B, and A * B means A is the father of B, which expression shows that C is the paternal grandmother of D?

A C % E * D

B C * E % D

C E % C * D

D D % E * C

✓ **Verified Answer:** (A)** — `C % E * D` **Detailed Step-by-Step Solution & Logical Derivation:** C % E: C is mother of E. E * D: E is father of D. So C is the mother of D's father, i.e., C is D's paternal grandmother.

Problem 41 • Infosys SE Question 41

CORE SE

ID: Q41 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
A man walks 12 m North, then 9 m East, then 8 m South, then 6 m West. How far is he from the starting point, and in which general direction?

Visual Schematic / Logical Deduction Diagram:

```
              (12m North)
          ┌──────────────────► (9m East)
          │                  │
          ▲                  ▼ (8m South)
      [START] ◄──────────────┘
                (6m West)
Net displacement: Δy = 12 - 8 = 4m North; Δx = 9 - 6 = 3m East
Shortest distance = √(3² + 4²) = 5 m.
```

Visual Schematic / Logical Deduction Diagram:

```
              (12m North)
          ┌──────────────────► (9m East)
          │                  │
          ▲                  ▼ (8m South)
      [START] ◄──────────────┘
                (6m West)
Net displacement: Δy = 12 - 8 = 4m North; Δx = 9 - 6 = 3m East
Shortest distance = √(3² + 4²) = 5 m.
```

Visual Schematic / Logical Deduction Diagram:

```
              (12m North)
          ┌──────────────────► (9m East)
          │                  │
          ▲                  ▼ (8m South)
      [START] ◄──────────────┘
                (6m West)
Net displacement: Δy = 12 - 8 = 4m North; Δx = 9 - 6 = 3m East
Shortest distance = √(3² + 4²) = 5 m.
```

Visual Schematic / Logical Deduction Diagram:

```
              (12m North)
          ┌──────────────────► (9m East)
          │                  │
          ▲                  ▼ (8m South)
      [START] ◄──────────────┘
                (6m West)
Net displacement: Δy = 12 - 8 = 4m North; Δx = 9 - 6 = 3m East
Shortest distance = √(3² + 4²) = 5 m.
```

A 5 m, North-East

B 5 m, North-West

C 7 m, North-East

D 5 m, South-East

✓ **Verified Answer:** (A)** — `5 m, North-East` **Detailed Step-by-Step Solution & Logical Derivation:** Net North-South displacement = 12−8 = 4 m North; net East-West = 9−6 = 3 m East. Distance = $\sqrt{(3^2+4^2)} = \sqrt{25} = 5$ m, in the general North-East direction.

Problem 42 • Infosys SE Question 42

CORE SE

`ID: Q42` `Source: Infosys PYQ` `Level: Infosys SE Standard`

(Infosys System Engineer ITM Mock Paper 3 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
A cyclist starts facing East. He turns 135° clockwise, then 90° anticlockwise, then 180° clockwise. Which direction is he facing now?

A North-West

B South-East

C South-West

D North-East

✓ **Verified Answer:** (A)** — `North-West` **Detailed Step-by-Step Solution & Logical Derivation:** Using bearings (clockwise from North, East=90°): 90°+135°=225°(SW); 225°−90°=135°(SE); 135°+180°=315°, which is North-West.

Problem 43 • Infosys SE Question 43

CORE SE

ID: Q43 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
Find the next term: 7, 26, 63, 124, 215, ?

A 342

B 344

C 341

D 343

✓ **Verified Answer:** (A)** — `342` **Detailed Step-by-Step Solution & Logical Derivation:** Each term is n^3-1 for n=2,3,4,5,6: 7,26,63,124,215; next (n=7): 343−1 = 342.

Problem 44 • Infosys SE Question 44

CORE SE

ID: Q44 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q14)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
Find the next term: 4, 5, 9, 18, 34, ?

A 55

B 59

C 63

D 57

✓ **Verified Answer:** (B)** — `59` **Detailed Step-by-Step Solution & Logical Derivation:** Differences are 1,4,9,16 (perfect squares); next difference is 25: 34+25 = 59.

Problem 45 • Infosys SE Question 45

CORE SE

ID: Q45 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q15)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
Find the next term: B, D, G, K, P, ?

A V

B U

C T

D W

✓ **Verified Answer:** (A)** — `V` **Detailed Step-by-Step Solution & Logical Derivation:** Differences between consecutive letters (by position) increase by 1 each time: +2,+3,+4,+5, next +6: P(16)+6 = 22 = V.

Problem 46 • Infosys SE Question 46

CORE SE

ID: Q46 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_04_page_01.png

Question Text:
Statements: Some pens are pencils. All pencils are erasers. No eraser is a sharpener.
Conclusions: I. No pencil is a sharpener. II. Some pens are erasers. III. Some erasers are not pens. IV. Either some pens are sharpeners or no pens are sharpeners.

A Only I and II follow

B I, II and IV follow

C All follow

D Only IV follows

✓ **Verified Answer:** (B)** — `I, II and IV follow` **Detailed Step-by-Step Solution & Logical Derivation:** Pencils⊆erasers and erasers∩sharpeners=∅, so pencils∩sharpeners=∅ (I follows). Some pens are pencils, and pencils⊆erasers, so some pens are erasers (II follows). III isn't guaranteed (no information ensures an eraser outside the pen-pencil overlap exists). For IV: we know the pen-pencils aren't sharpeners, but nothing is said about pens that aren't pencils, so neither "some pens are sharpeners" nor "no pens are sharpeners" can be individually proven — exactly the condition for the standard either-or rule, so IV follows.

Problem 47 • Infosys SE Question 47

CORE SE

ID: Q47 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_04_page_01.png

Question Text:
Statements: Only members are allowed entry. Some members are officers. All officers are trained.
Conclusions: I. Some trained persons are members. II. All those allowed entry are trained. III. Some members are trained.

A I and III only

B II and III only

C All follow

D None follows

✓ **Verified Answer:** (A)** — `I and III only` **Detailed Step-by-Step Solution & Logical Derivation:** "Only members are allowed entry" means allowed-entry⊆members. Some members are officers, and officers⊆trained, so those members are trained — this directly gives "some members are trained" (III) and, by simple conversion, "some trained are members" (I). But since only some members are officers/trained, we cannot say all those allowed entry are trained (II fails, as non-officer members may not be trained).

Problem 48 • Infosys SE Question 48

CORE SE

ID: Q48 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_04_page_01.png

Question Text:
Data Sufficiency. What is the value of x?
I. $x^2 - 5x + 6 = 0$
II. $x > 2$

A Statement I alone is sufficient, but II alone is not

B Statement II alone is sufficient, but I alone is not

C Both statements together are sufficient, but neither alone is

D Even both statements together are not sufficient

✓ **Verified Answer:** (C)** — `Both statements together are sufficient, but neither alone is` **Detailed Step-by-Step Solution & Logical Derivation:** I alone: $x^2-5x+6=0$ factors to $(x-2)(x-3)=0$, giving $x=2$ or $x=3$ — two solutions, insufficient. II alone: $x>2$ alone is insufficient. Together: only $x=3$ satisfies both $x>2$ and the equation — sufficient together.

Problem 49 • Infosys SE Question 49

CORE SE

ID: Q49 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_04_page_01.png

Question Text:
Data Sufficiency. In triangle ABC, what is the measure of angle B?
I. Angle A = 50°, and the triangle is isosceles with AB = AC.
II. Angle C = 65°.

A Statement I alone is sufficient, but II alone is not

B Statement II alone is sufficient, but I alone is not

C Both statements together are sufficient, but neither alone is

D Even both statements together are not sufficient

✓ **Verified Answer:** (A)** — `Statement I alone is sufficient, but II alone is not` **Detailed Step-by-Step Solution & Logical Derivation:** I alone: AB=AC means the angles opposite them are equal, so angle B = angle C; with angle A=50°, B+C=130°, and B=C gives B=65° — sufficient. II alone gives only one angle with no other relation — insufficient.

Problem 50 • Infosys SE Question 50

CORE SE

ID: Q50 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
In a code, each letter is first replaced by its position in the alphabet, and that number is then doubled and reduced by 1. Using this rule, CAT is coded as 5-1-39. What is the code for DOG?

A 7-29-13

B 8-29-13

C 7-30-13

D 7-29-14

✓ **Verified Answer:** (A)** — `7-29-13` **Detailed Step-by-Step Solution & Logical Derivation:** D→4→2(4)−1=7. O→15→2(15)−1=29. G→7→2(7)−1=13. Code: 7-29-13.

Problem 51 • Infosys SE Question 51

CORE SE

ID: Q51 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
In a code: "alpha beta gamma delta" = "zx qr mn kl", "beta gamma epsilon" = "qr mn vw", "gamma delta zeta" = "mn kl tu", "epsilon zeta eta" = "vw tu yz", "alpha eta theta" = "zx yz op". What is the code for "theta beta zeta"?

A op qr tu

B op qr vw

C zx qr tu

D op mn tu

✓ **Verified Answer:** (A)** — `op qr tu` **Detailed Step-by-Step Solution & Logical Derivation:** Comparing statements pairwise for common words and codes: shines=pu, gamma=mn, bright=mo (from statements 1&3), giving delta=kl (from statements 1&3's remaining elements), light=ta, zeta=tu, epsilon=vw, eta=yz, alpha=zx, theta=op. So "theta beta zeta" = op qr tu.

Problem 52 • Infosys SE Question 52

CORE SE

ID: Q52 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
A cube of side 6 units is painted red on all faces and then cut into 1-unit cubes. How many of the small cubes have at least 2 painted faces?

Visual Schematic / Logical Deduction Diagram:

```
6 × 6 × 6 Cube (216 unit cubes)
Painted RED on all faces.
Cubes with NO paint = (6 - 2)³ = 4³ = 64 cubes.
Cubes with at least 1 face painted = 216 - 64 = 152 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
6 × 6 × 6 Cube (216 unit cubes)
Painted RED on all faces.
Cubes with NO paint = (6 - 2)³ = 4³ = 64 cubes.
Cubes with at least 1 face painted = 216 - 64 = 152 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
6 × 6 × 6 Cube (216 unit cubes)
Painted RED on all faces.
Cubes with NO paint = (6 - 2)³ = 4³ = 64 cubes.
Cubes with at least 1 face painted = 216 - 64 = 152 cubes.
```

Visual Schematic / Logical Deduction Diagram:

```
6 × 6 × 6 Cube (216 unit cubes)
Painted RED on all faces.
Cubes with NO paint = (6 - 2)³ = 4³ = 64 cubes.
Cubes with at least 1 face painted = 216 - 64 = 152 cubes.
```

- **A** 56
- **B** 48
- **C** 64
- **D** 60

✓ **Verified Answer:** (A)** — `56` **Detailed Step-by-Step Solution & Logical Derivation:** For an n×n×n cube, cubes with exactly 2 painted faces (edges) = 12(n−2); with exactly 3 (corners) = 8. For n=6: edges=12×4=48, corners=8, so at least 2 = 48+8 = 56.

Problem 53 • Infosys SE Question 53

CORE SE

ID: Q53 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
How many squares (of all sizes) are there in a 4×4 grid of unit squares?

A 30

B 25

C 36

D 20

✓ **Verified Answer:** (A)** — `30` **Detailed Step-by-Step Solution & Logical Derivation:** Squares of size k in an n×n grid = $(n-k+1)^2$. For n=4: k=1→16, k=2→9, k=3→4, k=4→1. Total = 16+9+4+1 = 30.

Problem 54 • Infosys SE Question 54

CORE SE

ID: Q54 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
P is the father of Q. Q is the brother of R. S is the mother of R. T is the husband of S. How is P related to T?

A Father

B P and T are the same person

C Brother-in-law

D Cannot be determined

✓ **Verified Answer:** (B)** — `P and T are the same person` **Detailed Step-by-Step Solution & Logical Derivation:** Q and R are siblings (Q is R's brother), so S (R's mother) is also Q's mother. P is Q's father, so P and S are Q's parents — meaning P is S's husband. Since T is defined as S's husband, P and T are the same person.

Problem 55 • Infosys SE Question 55

CORE SE

ID: Q55 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
If A + B means A is the father of B, A − B means A is the mother of B, and A × B means A is the son of B, which expression shows that M is the maternal grandfather of N?

A M + Q − N

B M − Q + N

C Q + M − N

D M × Q − N

✓ **Verified Answer:** (A)** — `M + Q − N` **Detailed Step-by-Step Solution & Logical Derivation:** M+Q: M is father of Q. Q−N: Q is mother of N. So M is father of Q, who is N's mother — M is N's maternal grandfather.

Problem 56 • Infosys SE Question 56

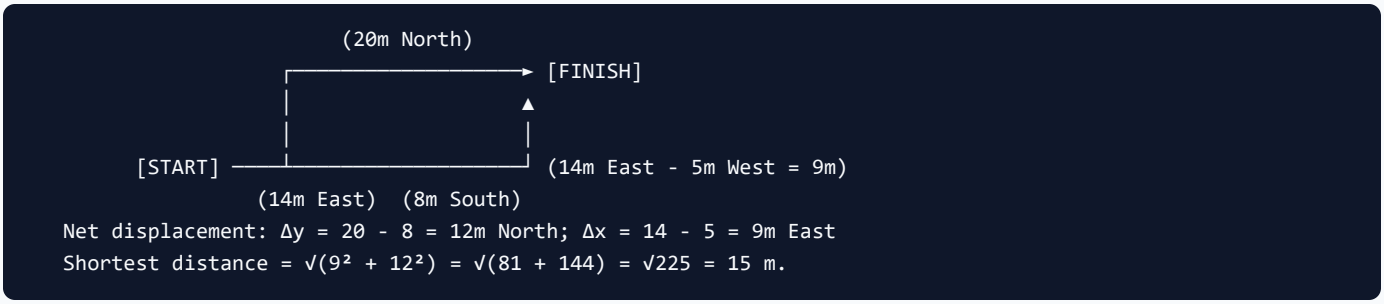
CORE SE

ID: Q56 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

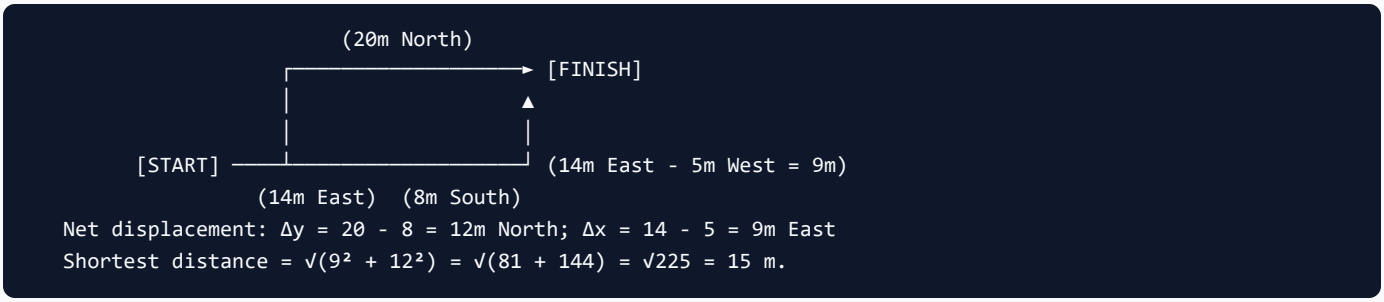
Question Text:
A man walks 14 m East, then 20 m North, then 5 m West, then 8 m South. How far is he from the starting point, and in which direction?

Visual Schematic / Logical Deduction Diagram:



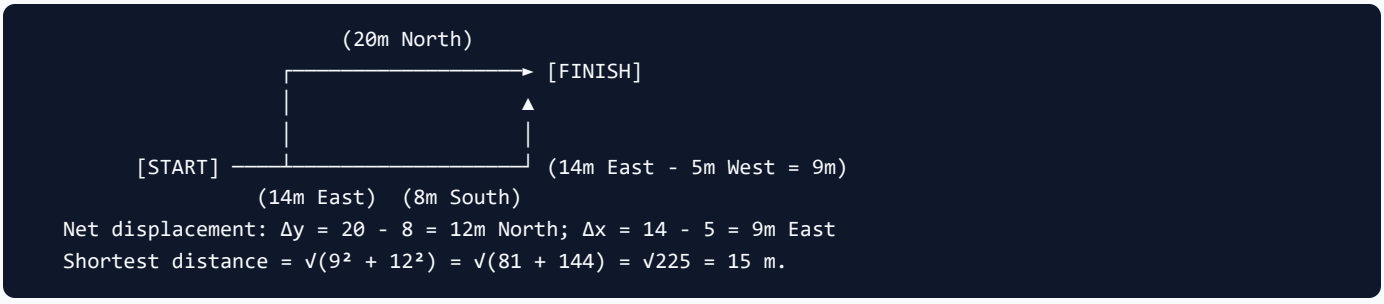
Net displacement: $\Delta y = 20 - 8 = 12$m North; $\Delta x = 14 - 5 = 9$m East
Shortest distance = $\sqrt{(9^2 + 12^2)} = \sqrt{(81 + 144)} = \sqrt{225} = 15$ m.

Visual Schematic / Logical Deduction Diagram:



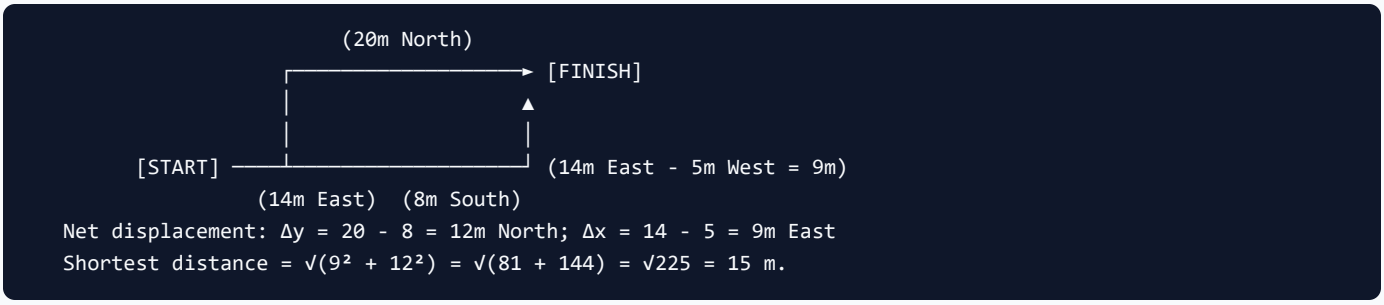
Net displacement: $\Delta y = 20 - 8 = 12$m North; $\Delta x = 14 - 5 = 9$m East
Shortest distance = $\sqrt{(9^2 + 12^2)} = \sqrt{(81 + 144)} = \sqrt{225} = 15$ m.

Visual Schematic / Logical Deduction Diagram:



Net displacement: $\Delta y = 20 - 8 = 12$m North; $\Delta x = 14 - 5 = 9$m East
Shortest distance = $\sqrt{(9^2 + 12^2)} = \sqrt{(81 + 144)} = \sqrt{225} = 15$ m.

Visual Schematic / Logical Deduction Diagram:



Net displacement: $\Delta y = 20 - 8 = 12$m North; $\Delta x = 14 - 5 = 9$m East
Shortest distance = $\sqrt{(9^2 + 12^2)} = \sqrt{(81 + 144)} = \sqrt{225} = 15$ m.

A 15 m, North-East

B 15 m, North-West

C 13 m, North-East

D 17 m, North-East

✓ **Verified Answer:** (A)** — `15 m, North-East` **Detailed Step-by-Step Solution & Logical Derivation:** Net East-West = 14−5 = 9 m East; net North-South = 20−8 = 12 m North. Distance = $\sqrt{(9^2+12^2)}$ = √225 = 15 m, direction North-East.

Problem 57 • Infosys SE Question 57

CORE SE

ID: Q57 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
A ship sails 10 km from port on a bearing of 030° (N30°E), then sails 10 km on a bearing of 150° (S30°E). What is the ship's final position relative to the port?

A 10 km due East (bearing 090°)

B 10 km due North

C 17.3 km, bearing 060°

D 10 km, bearing 120°

✓ **Verified Answer:** (A)** — `10 km due East (bearing 090°)` **Detailed Step-by-Step Solution & Logical Derivation:** Leg 1 (bearing 030°, 10 km): East = 10 sin30° = 5, North = 10 cos30° = 5√3. Leg 2 (bearing 150°, 10 km): East = 10 sin150° = 5, North = 10 cos150° = −5√3. Totals: East = 5+5 = 10, North = 5√3−5√3 = 0. Final position: 10 km due East (bearing 090°).

Problem 58 • Infosys SE Question 58

CORE SE

ID: Q58 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
Find the next term: 2, 5, 12, 27, 58, ?

A 121

B 125

C 115

D 118

✓ **Verified Answer:** (A)** — `121` **Detailed Step-by-Step Solution & Logical Derivation:** Differences are 3, 7, 15, 31 — each is 2^n-1 (n=2,3,4,5); next difference is $2^6-1=63$. 58+63 = 121.

Problem 59 • Infosys SE Question 59

CORE SE

ID: Q59 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q14)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
Find the next term: 1, 3, 7, 17, 41, ?

A 99

B 95

C 101

D 89

✓ **Verified Answer:** (A)** — `99` **Detailed Step-by-Step Solution & Logical Derivation:** Each term follows a(n) = 2×a(n−1) + a(n−2): 2×17+7=41 ✓; next = 2×41+17 = 99.

Problem 60 • Infosys SE Question 60

CORE SE

ID: Q60 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q15)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
Find the next term: Y, V, R, M, ?

A G

B H

C F

D I

✓ **Verified Answer:** (A)** — `G` **Detailed Step-by-Step Solution & Logical Derivation:** Gaps between letters increase by 1 each time: −3, −4, −5, next −6. M(13)−6 = 7 = G.

Problem 61 • Infosys SE Question 61

CORE SE

ID: Q61 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q01)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_05_page_01.png

Question Text:
Statements: All toys are games. All games are activities. Some crafts are not activities. No craft is a hobby.
Conclusions: I. All toys are activities. II. Some crafts are not hobbies. III. No toy is a craft.

A Only I follows

B I and II follow

C I, II and III follow

D None follows

✓ **Verified Answer:** (B)** — `I and II follow` **Detailed Step-by-Step Solution & Logical Derivation:** Toys⊆games⊆activities, so all toys are activities (I follows). "Some crafts are not activities" confirms crafts exist, and since no craft is a hobby, those crafts are also not hobbies — "some crafts are not hobbies" follows (II). III is not guaranteed, since nothing links toys specifically to crafts.

Problem 62 • Infosys SE Question 62

CORE SE

ID: Q62 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q02)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_05_page_01.png

Question Text:
Statements: All researchers are scholars. No scholar is lazy. Some lazy people are students. All students are young.
Conclusions: I. No researcher is lazy. II. Some students are not scholars. III. Some young people are lazy.

A Only I and II follow

B Only II and III follow

C All follow

D None follows

✓ **Verified Answer:** (C)** — `All follow` **Detailed Step-by-Step Solution & Logical Derivation:** Researchers⊆scholars, and scholars∩lazy=∅, so researchers∩lazy=∅ (I). Some lazy are students, and since scholars∩lazy=∅, those lazy-students are not scholars — "some students are not scholars" (II). Those same lazy-students are also students, and all students are young, so those lazy-students are young — "some young people are lazy" (III). All three follow.

Problem 63 • Infosys SE Question 63

CORE SE

ID: Q63 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q03)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_05_page_01.png

Question Text:
Data Sufficiency. If x and y are positive integers, what is the remainder when x is divided by y?
I. x = 5y + 3
II. y = 7

A Statement I alone is sufficient, but II alone is not

B Statement II alone is sufficient, but I alone is not

C Both statements together are sufficient, but neither alone is

D Even both statements together are not sufficient

✓ **Verified Answer:** (C)** — `Both statements together are sufficient, but neither alone is` **Detailed Step-by-Step Solution & Logical Derivation:** I alone: x=5y+3 only gives remainder 3 if y>3; for small y (e.g., y=2, x=13, 13÷2 leaves remainder 1, not 3) it fails — insufficient alone. II alone: y=7 with x unknown is insufficient. Together: x=5(7)+3=38, and 38÷7 leaves remainder 3 (valid since 3<7) — sufficient together.

Problem 64 • Infosys SE Question 64

CORE SE

ID: Q64 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q04)
Section: Section 1: Reasoning Ability | **Page:** 1
Visual Reference Plate: paper_05_page_01.png

Question Text:
Data Sufficiency. In how many days can A and B complete a job together?
I. A is twice as efficient as B.
II. A alone can complete the job in 12 days.

A Statement I alone is sufficient, but II alone is not

B Statement II alone is sufficient, but I alone is not

C Both statements together are sufficient, but neither alone is

D Even both statements together are not sufficient

✓ **Verified Answer:** (C)** — `Both statements together are sufficient, but neither alone is` **Detailed Step-by-Step Solution & Logical Derivation:** II alone gives only A's rate (1/12), with B's rate unknown. I alone gives only a ratio (2:1), with no absolute rate. Together: A's rate=1/12, B's rate=1/24 (half of A's), combined rate=1/12+1/24=3/24=1/8, so together they take 8 days — sufficient together.

Problem 65 • Infosys SE Question 65

CORE SE

ID: Q65 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q05)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
In a code, letters in odd positions of a word are shifted 2 places forward in the alphabet, and letters in even positions are shifted 1 place forward. Using this rule, SYSTEM is coded as UZUUGN. What is the code for SERVER?

A UFTWGS

B UFTWFS

C UGTWGS

D UFUWGS

✓ **Verified Answer:** (A)** — `UFTWGS` **Detailed Step-by-Step Solution & Logical Derivation:** S(odd,+2)=U. E(even,+1)=F. R(odd,+2)=T. V(even,+1)=W. E(odd,+2)=G. R(even,+1)=S. Code: UFTWGS.

Problem 66 • Infosys SE Question 66

CORE SE

ID: Q66 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q06)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
In a code: "data packet loss" = "zt rp qm", "loss causes retransmission" = "qm xy lp", "retransmission increases latency" = "lp wv ou", "latency affects data" = "ou nk zt", "packet causes latency" = "rp xy ou". What is the code for "increases affects packet"?

A wv nk rp

B wv nk qm

C ou nk rp

D wv xy rp

✓ **Verified Answer:** (A)** — `wv nk rp` **Detailed Step-by-Step Solution & Logical Derivation:** Pairwise comparison of the 5 statements resolves every word: data=zt, packet=rp, loss=qm, causes=xy, retransmission=lp, increases=wv, latency=ou, affects=nk. So "increases affects packet" = wv nk rp.

Problem 67 • Infosys SE Question 67

CORE SE

ID: Q67 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q07)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
A rectangular cuboid is cut by 2 parallel cuts along its length, 1 cut along its breadth, and 3 cuts along its height (each cut perpendicular to that dimension), producing smaller cuboids from an originally fully-painted surface. How many of the resulting small cuboids have exactly 1 painted face?

A 4

B 6

C 8

D 2

✓ **Verified Answer:** (A)** — `4` **Detailed Step-by-Step Solution & Logical Derivation:** With 2 cuts in length (3 pieces), 1 cut in breadth (2 pieces), 3 cuts in height (4 pieces), every piece touches an outer breadth-face (since there's no "middle" breadth layer). A piece has exactly 1 painted face only if it is interior in BOTH length and height (contributing 0 from each) while contributing exactly 1 from breadth (which is unavoidable). Interior length positions = 1 (out of 3), interior height positions = 2 (out of 4), and both breadth pieces qualify: 1×2×2 = 4.

Problem 68 • Infosys SE Question 68

CORE SE

ID: Q68 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q08)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
How many rectangles (of all sizes, including squares) are there in a 4×4 grid of unit squares?

A 100

B 90

C 81

D 110

✓ **Verified Answer:** (A)** — `100` **Detailed Step-by-Step Solution & Logical Derivation:** Choosing 2 of the 5 horizontal lines and 2 of the 5 vertical lines of a 4×4 grid gives a rectangle: C(5,2)×C(5,2) = 10×10 = 100.

Problem 69 • Infosys SE Question 69

CORE SE

ID: Q69 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q09)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
P and Q are a married couple. R is the son of P and Q. S is the daughter of P and Q. R is married to T. U is the son of R and T. V is the daughter of S. How is U related to V?

A Cousins

B Siblings

C Nephew and Niece

D Uncle and Niece

✓ **Verified Answer:** (A)** — `Cousins` **Detailed Step-by-Step Solution & Logical Derivation:** R and S are siblings (children of P and Q). U is R's son; V is S's daughter. Since their parents (R and S) are siblings, U and V are cousins.

Problem 70 • Infosys SE Question 70

CORE SE

ID: Q70 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q10)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
If A + B means A is the father of B, A − B means A is the mother of B, A × B means A is the son of B, and A ÷ B means A is the wife of B, which expression shows that M is the father-in-law of N (N being male)?

A M + Q ÷ N

B M ÷ Q + N

C Q + M ÷ N

D M × Q ÷ N

✓ **Verified Answer:** (A)** — `M + Q ÷ N` **Detailed Step-by-Step Solution & Logical Derivation:** M+Q: M is father of Q. Q÷N: Q is wife of N. So M is father of N's wife — M is N's father-in-law.

Problem 71 • Infosys SE Question 71

CORE SE

ID: Q71 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q11)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
A man starts from point O and walks 10 m North-East, then 10 m North-West, then 10 m South-West.
How far is he from O, and in which direction?

Visual Schematic / Logical Deduction Diagram:

```
                (10m NW)
            ┌──────────────► [Point 2]
            │               /
(10m NE)    ▲              / (10m SW)
            │             ▼
        [Point O] ──► [FINISH]
Net position: returns to baseline axis at distance 10 m from O.
```

Visual Schematic / Logical Deduction Diagram:


```
                (10m NW)
            ┌──────────────► [Point 2]
            │               /
(10m NE)    ▲              / (10m SW)
            │             ▼
        [Point O] ──► [FINISH]
Net position: returns to baseline axis at distance 10 m from O.
```


Visual Schematic / Logical Deduction Diagram:


```
                (10m NW)
            ┌──────────────► [Point 2]
            │               /
(10m NE)    ▲              / (10m SW)
            │             ▼
        [Point O] ──► [FINISH]
Net position: returns to baseline axis at distance 10 m from O.
```


Visual Schematic / Logical Deduction Diagram:


```
                (10m NW)
            ┌──────────────► [Point 2]
            │               /
(10m NE)    ▲              / (10m SW)
            │             ▼
        [Point O] ──► [FINISH]
Net position: returns to baseline axis at distance 10 m from O.
```


A 10 m, North-West

B 10 m, South-West

C 14.14 m, North-West

D 5 m, North-West

✓ **Verified Answer:** (A)** — `10 m, North-West` **Detailed Step-by-Step Solution & Logical Derivation:** Decomposing each 10 m leg into East/North components (each at 45°, so ±5√2 in each axis): totals work out to East=−5√2, North=+5√2, giving a resultant of √[(5√2)²+(5√2)²]=√100=10 m, in the North-West direction.

Problem 72 • Infosys SE Question 72

CORE SE

`ID: Q72` `Source: Infosys PYQ` `Level: Infosys SE Standard`

(Infosys System Engineer ITM Mock Paper 5 - Q12)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
A man is initially facing a bearing of 040°. After making exactly 2 equal clockwise turns, he ends up facing a bearing of 310°. What was the magnitude of each turn?

A 135°

B 142.5°

C 120°

D 150°

✓ **Verified Answer:** (A)** — `135°` **Detailed Step-by-Step Solution & Logical Derivation:** Total clockwise rotation needed = 310° −40° = 270°. Split equally over 2 turns: 270/2 = 135° each (verified: 40°+135°=175°, 175°+135°=310° ✓).

Problem 73 • Infosys SE Question 73

CORE SE

ID: Q73 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q13)
Section: Section 1: Reasoning Ability | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
Find the next term: 5, 14, 41, 122, 365, ?

A 1094

B 1095

C 1093

D 1096

✓ **Verified Answer:** (A)** — `1094` **Detailed Step-by-Step Solution & Logical Derivation:** Each term follows a(n) = 3×a(n−1) − 1: 122×3−1=365 ✓; next = 365×3−1 = 1094.